Introduction to Analysis II

Lecture notes, Chapters 6–10

Contents

Chapter 6. Sequences and Series of Functions 3

6.1 Uniform convergence 3

6.2 Differentiation and integration of limits 4

6.3 Double sequences and double series 5

6.4 Power series 6

6.5 Trigonometric series 7

6.6 Selected exercises 9

Chapter 7. Function Spaces 12

7.1 The space $C(X)$ 12

7.2 Sequence spaces 13

7.3 Improper integrals 14

7.4 Spaces of integrable functions 15

7.5 Selected exercises 16

Chapter 8. Functions Defined by Integrals 19

8.1 Integrals depending on a parameter 19

8.2 The gamma function 20

8.3 Selected exercises 21

Chapter 9. Fourier Series 24

9.1 Fourier coefficients 24

9.2 Pointwise convergence 25

9.3 Fejér's theorem 26

9.4 Differentiation and integration of Fourier series 28

9.5 Selected exercises 29

Chapter 10. The Lebesgue Integral **32**

10.1 Measurable sets and Lebesgue measure **32**

10.2 Measurable functions and the integral **33**

10.3 Convergence theorems **35**

10.4 The spaces L^1 and L^2 **37**

10.5 The Poisson summation formula **39**

Chapter 6. Sequences and Series of Functions

6.1 Uniform convergence

Motivation. Functions are often defined as limits: power series, Fourier series, solutions of differential equations. Pointwise limits can destroy continuity (x^n on $[0,1]$), so we need a stronger notion. Uniform convergence is the one that preserves continuity, and the Weierstrass M-test is the usual way to verify it.

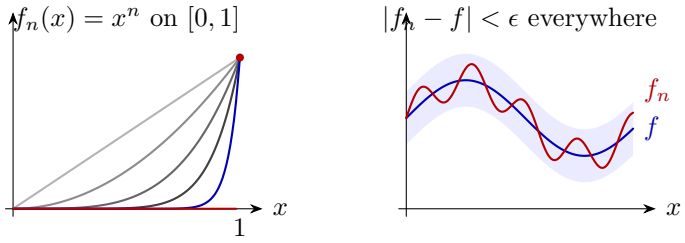


Figure. *Left: x^n converges pointwise on $[0,1]$ to a discontinuous limit (0 for $x<1$, 1 at $x=1$), so the convergence cannot be uniform. Right: uniform convergence means that for large n the whole graph of f_n lies in the ϵ-band around f.*

Let $X \subset \mathbb{R}^n$ and $f_n, f : X \to \mathbb{R}^m$ (or $\mathbb{C}$). $f_n \to f$ **pointwise** if $f_n(x) \to f(x)$ for each x, and **uniformly** if $\|f_n - f\|_{\sup} = \sup_{x\in X}\|f_n(x) - f(x)\| \to 0$.

Example 6.1

$x^n \to 0$ pointwise on $[0,1)$ but not uniformly, since $\sup_{[0,1)} x^n = 1$; the limit of x^n on $[0,1]$ is discontinuous. $f_n(x) = nx(1-x^2)^n$ tends to 0 pointwise on $[0,1]$, while $\int_0^1 f_n = \frac{n}{2n+2} \to \frac{1}{2}$.

Theorem 6.1 (6.1.1)

If continuous $f_n : X \to Y$ converge uniformly to f, then f is continuous.

Proof. Given x_0 and ϵ, choose N with $\|f_N - f\|_{\sup} < \epsilon/3$ and δ with $\|f_N(x) - f_N(x_0)\| < \epsilon/3$ for $\|x - x_0\| < \delta$. Then

$$\|f(x) - f(x_0)\| \le \|f(x) - f_N(x)\| + \|f_N(x) - f_N(x_0)\| + \|f_N(x_0) - f(x_0)\| < \epsilon.$$

■

Theorem 6.2 (6.1.2, uniform Cauchy criterion)

(f_n) converges uniformly on X iff for every $\epsilon > 0$ there is N with $\|f_m(x) - f_n(x)\| < \epsilon$ for all $m, n \ge N$ and $x \in X$.

Proof. Necessity is the triangle inequality. Conversely, each $(f_n(x))$ is Cauchy, so converges to some $f(x)$ (Theorem 2.14 of Volume I); letting $m \to \infty$ in $\|f_m(x) - f_n(x)\| < \epsilon$ gives $\|f(x) - f_n(x)\| \le \epsilon$ for all x and $n \ge N$. ■

Corollary 6.3 (6.1.3, Weierstrass M-test)

If $|f_n(x)| \le M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly on X.

Proof. For $n > m$, $|\sum_{k=m+1}^{n} f_k(x)| \le \sum_{k=m+1}^{n} M_k$, which is small uniformly in x; apply Theorem 6.2 to the partial sums. ■

Example 6.2

$\sum 2^{-n}\cos(3^n x)$ converges uniformly, so it is continuous. Weierstrass's function $\sum a^n \cos(b^n \pi x)$ ($0 < a < 1$, $ab > 1 + 3\pi/2$, b odd) is continuous but nowhere differentiable. Also, there is a continuous curve filling a square (Peano), obtained as a uniform limit.

6.2 Differentiation and integration of limits

Motivation. When can we exchange a limit with an integral or a derivative? Uniform convergence allows the first; the second needs uniform convergence of the derivatives. These are the theorems that justify term-by-term manipulation of series.

Theorem 6.4 (6.2.1)

Let $f_n : [a,b] \to \mathbb{R}$ satisfy:
(a) each f_n is differentiable;
(b) $(f_n(x_0))$ converges for some $x_0 \in [a,b]$;
(c) (f_n') converges uniformly. Then (f_n) converges uniformly to a differentiable f, and $f'(x) = \lim f_n'(x)$.

Proof. By the mean value theorem applied to $f_m - f_n$,

$$|(f_m - f_n)(x) - (f_m - f_n)(t)| \le |x - t|\, \|f_m' - f_n'\|_{\sup}. \qquad (*)$$

With $t = x_0$ this shows that (f_n) is uniformly Cauchy, hence converges uniformly to some f. Fix x and set $\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}$, $\phi(t) = \frac{f(t) - f(x)}{t - x}$ for $t \ne x$. By $(*)$, (ϕ_n) is uniformly Cauchy on $[a,b] - \{x\}$, so $\phi_n \to \phi$ uniformly there. Let $g = \lim f_n'$. Given ϵ, choose N with $|\phi_N - \phi| < \epsilon$ on $[a,b] - \{x\}$ and $|f_N'(x) - g(x)| < \epsilon$, and δ with $|\phi_N(t) - f_N'(x)| < \epsilon$ for $0 < |t - x| < \delta$. Then $|\phi(t) - g(x)| < 3\epsilon$ for such t, so $f'(x) = g(x)$. ■

Uniform convergence of (f_n) alone is not enough: $\sin(nx)/\sqrt{n} \to 0$ uniformly, but the derivatives $\sqrt{n}\cos(nx)$ diverge.

Proposition 6.5 (6.2.2)

If Riemann integrable $f_n : [a,b] \to \mathbb{R}$ converge uniformly to f, then f is integrable and $\int_a^b f = \lim \int_a^b f_n$.

Proof. Let $\epsilon_n = \|f_n - f\|_{\sup}$. From $f_n - \epsilon_n \le f \le f_n + \epsilon_n$, the upper and lower integrals of f lie within $\epsilon_n(b-a)$ of $\int f_n$; so they differ by at most $2\epsilon_n(b-a) \to 0$, and $|\int f - \int f_n| \le \epsilon_n(b-a)$. ■

Example 6.3

Integrating $\frac{1}{1+t^2} = \sum_{n\geq 0}(-1)^n t^{2n}$ (uniform on $[-r,r]$, $r<1$) gives $\arctan x = \sum(-1)^n x^{2n+1}/(2n+1)$ for $|x|<1$; similarly $\log(1+x) = \sum(-1)^{n+1}x^n/n$. If $a_n > 0$ decrease and $\sum a_n < \infty$, then $na_n \to 0$: $\frac{n}{2}a_n \leq a_{\lceil n/2\rceil} + \cdots + a_n \to 0$.

6.3 Double sequences and double series

Motivation. Iterated limits and sums cannot always be exchanged. This section gives conditions, absolute convergence above all, under which the order of summation does not matter; they are needed for products of series and for power series in two variables.

Proposition 6.6 (6.3.1)

Let (a_{mn}) be a double sequence such that (a) $a_{mn} \to b_m$ as $n \to \infty$ for each m;
(b) $a_{mn} \to c_n$ as $m \to \infty$ for each n;
(c) the convergence in (b) is uniform in n: for every ϵ there is N with $|a_{mn}-c_n|<\epsilon$ for $m \geq N$ and all n. Then (b_m) and (c_n) converge to the same limit: $\lim_m \lim_n a_{mn} = \lim_n \lim_m a_{mn}$.

Proof. Regard $f_m(n) = a_{mn}$ as functions on $X = \{1, 1/2, 1/3, \dots\} \cup \{0\}$ (with $n \leftrightarrow 1/n$), extended to 0 by $f_m(0) = b_m$; each f_m is continuous by (a), and by (c) and Theorem 6.2 the f_m converge uniformly on X (at 0 too, since $|b_m - b_k| \leq \sup_n |a_{mn} - a_{kn}|$). The limit is continuous (Theorem 6.1), with value c_n at $1/n$ and $\lim b_m$ at 0; continuity at 0 gives $\lim_n c_n = \lim_m b_m$. ■

Proposition 6.7 (6.3.2)

If continuous $f_n : X \to Y$ converge uniformly to f and $x_n \to x$ in X, then $f_n(x_n) \to f(x)$.

Proof. $\|f_n(x_n) - f(x)\| \leq \|f_n - f\|_{\sup} + \|f(x_n) - f(x)\|$, and f is continuous (Theorem 6.1). ■

With pointwise convergence this fails: $f_n(x) = x^n$, $x_n = 1 - 1/n$, gives $f_n(x_n) \to 1/e$ while $f(1) = 1$.

Proposition 6.8 (6.3.3)

If $\sum_n |a_{mn}| = b_m < \infty$ for each m and $\sum_m b_m < \infty$, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$, all series converging absolutely.

Proof. Each $\sum_m |a_{mn}| \leq \sum_m b_m$, so the inner sums converge absolutely.

Let $S_{MN} = \sum_{m\leq M} \sum_{n \leq N} a_{mn}$. As $N \to \infty$, $S_{MN} \to \sum_{m\leq M}\sum_n a_{mn}$; as $M \to \infty$, $S_{MN} \to \sum_{n\leq N}\sum_m a_{mn}$, uniformly in N, since

$$\left| S_{MN} - \sum_{n\leq N}\sum_m a_{mn} \right| \leq \sum_{m>M} b_m.$$

Proposition 6.6 gives the equality of the iterated limits. ■

Proposition 6.9 (6.3.4)

If $a_{mn} \geq 0$ for all m, n, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$ in $[0, \infty]$.

Proof. For all M, N, $\sum_{m\leq M}\sum_{n\leq N} a_{mn} \leq \sum_{n\leq N}\sum_m a_{mn} \leq \sum_n \sum_m a_{mn}$. Letting $N \to \infty$ and then $M \to \infty$ gives $\leq$; the reverse is symmetric. ■

Both sides also equal $\sup_F \sum_{(m,n)\in F} a_{mn}$ over finite sets F, and any enumeration of the double series of nonnegative terms has this sum (Proposition 2.16 of Volume I). For $(a_k)_{k\in\mathbb{Z}}$, $\sum_{k=-\infty}^{\infty} a_k$ means $\lim_{m,n\to\infty} \sum_{k=-m}^{n} a_k$ when this double limit exists.

Corollary 6.10 (6.3.5)

If $a_k \geq 0$ for all $k \in \mathbb{Z}$, then $\lim_n \sum_{-n}^{n} a_k = \lim_n \lim_m \sum_{-m}^{n} a_k = \lim_m \lim_n \sum_{-m}^{n} a_k$ in $[0, \infty]$.

Proof. All three equal $\sup_{m,n} \sum_{-m}^{n} a_k$, since the partial sums increase in both m and n. ■

Corollary 6.11 (6.3.6)

If $\sum_{k\in\mathbb{Z}} |a_k| < \infty$, the three limits in Corollary 6.10 exist and coincide.

Proof. Apply Corollary 6.10 to $a_k^{\pm} = \max(\pm a_k, 0)$, whose sums are finite, and subtract. ■

6.4 Power series

Motivation. Power series are the most important example of uniform convergence. They converge uniformly on smaller discs, can be differentiated term by term, and therefore define smooth functions such as e^x, $\sin x$, $\cos x$ rigorously.

Theorem 6.12 (6.4.1, Cauchy–Hadamard)

Let $\lambda = \limsup |a_n|^{1/n}$ and $R = 1/\lambda$ ($R = \infty$ if $\lambda = 0$, $R = 0$ if $\lambda = \infty$). Then $\sum a_n x^n$ converges absolutely for $|x| < R$ and diverges for $|x| > R$; for every $0 < r < R$ it converges uniformly on $[-r, r]$. Its sum f is differentiable on $(-R, R)$ with $f'(x) = \sum n a_n x^{n-1}$, and hence C^∞, with $a_n = f^{(n)}(0)/n!$.

Proof. $\limsup |a_n x^n|^{1/n} = \lambda |x|$; apply the root test (Proposition 2.19 of Volume I). On $[-r, r]$ use the M-test with $M_n = |a_n| r^n$, summable as $r < R$. Since $\limsup |n a_n|^{1/n} = \lambda$ ($n^{1/n} \to 1$), the derived series has the same radius and converges uniformly on $[-r, r]$; Theorem 6.4 gives $f' = \sum n a_n x^{n-1}$. Iterate. ■

Theorem 6.13 (6.4.2, re-expansion)

If $f(x) = \sum a_n x^n$ on $(-R, R)$ and $|c| < R$, then $f(x) = \sum_n \frac{f^{(n)}(c)}{n!}(x-c)^n$ for $|x - c| < R - |c|$.

Proof. $f(x) = \sum_n a_n (c + (x-c))^n = \sum_n \sum_{k\leq n} a_n \binom{n}{k} c^{n-k} (x-c)^k$. Since $\sum_n \sum_k |a_n| \binom{n}{k} |c|^{n-k} |x-c|^k = \sum_n |a_n| (|c| + |x-c|)^n < \infty$, Proposition 6.8 allows the order of summation to be exchanged: $f(x) = \sum_k (\sum_n \binom{n}{k} a_n c^{n-k})(x-c)^k$, and the coefficient of $(x-c)^k$ is $f^{(k)}(c)/k!$ by termwise differentiation (Theorem 6.12). ■

So a power series is **analytic** (locally a power series) throughout its interval of convergence. The **Cauchy product** of $\sum a_n$ and $\sum b_n$ is $\sum c_n$, $c_n = \sum_{k=0}^{n} a_k b_{n-k}$.

Proposition 6.14 (6.4.3, Mertens)

If $\sum a_n = A$ converges absolutely and $\sum b_n = B$, then $\sum c_n = AB$.

Proof. With partial sums A_n, B_n, C_n and $\beta_n = B_n - B$, $C_n = \sum_{k\le n} a_k B_{n-k} = A_n B + \gamma_n$, $\gamma_n = \sum_{k\le n} a_k \beta_{n-k}$. Let $\alpha = \sum |a_k|$ and $\epsilon > 0$; choose N with $|\beta_m| < \epsilon$ for $m \ge N$. Then $|\gamma_n| \le \epsilon\alpha + \sum_{n-N<k\le n} |a_k||\beta_{n-k}|$, and the last sum tends to 0 as $n \to \infty$ because $a_k \to 0$ and the β's are bounded. So $\gamma_n \to 0$ and $C_n \to AB$. ∎

Theorem 6.15 (6.4.4, Abel)

Let $f(x) = \sum a_n x^n$ on $(-R, R)$. If $\sum a_n R^n$ converges, then $\lim_{x\to R^-} f(x) = \sum a_n R^n$.

Proof. Replacing a_n by $a_n R^n$ we may take $R = 1$; subtracting a constant, assume $s = \sum a_n = 0$. With partial sums $s_n \to 0$, summation by parts gives $\sum_{n\le N} a_n x^n = (1-x)\sum_{n<N} s_n x^n + s_N x^N$, so $f(x) = (1-x)\sum s_n x^n$ for $|x| < 1$. Given ϵ, choose M with $|s_n| < \epsilon$ for $n \ge M$; then $|f(x)| \le (1-x)\sum_{n<M} |s_n| + \epsilon$ for $0 < x < 1$, which is $< 2\epsilon$ for x near 1. ∎

Corollary 6.16 (6.4.5)

If $\sum a_n = A$, $\sum b_n = B$ and their Cauchy product $\sum c_n = C$ all converge, then $C = AB$.

Proof. For $|x| < 1$ the series $\sum a_n x^n$, $\sum b_n x^n$ converge absolutely, so $\sum c_n x^n = (\sum a_n x^n)(\sum b_n x^n)$ by Proposition 6.14. Let $x \to 1^-$ and use Theorem 6.15 three times. ∎

Example 6.4

Abel's theorem gives $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \cdots$ and $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ from Example 6.3. Defining $e^x = \sum x^n/n!$, Proposition 6.14 gives $e^{x+y} = e^x e^y$; with $\sin, \cos$ defined by their series, $(\sin)' = \cos$, $(\cos)' = -\sin$, and $\sin^2 + \cos^2 = 1$ follow by termwise differentiation.

6.5 Trigonometric series

Motivation. Fourier analysis represents a periodic function as a sum of sines and cosines. Here we prove the basic properties of trigonometric series that converge uniformly, as preparation for Chapter 9.

A complex vector space V with a map $\langle \cdot, \cdot \rangle$ that is linear in the first variable, satisfies $\langle y, x\rangle = \overline{\langle x, y\rangle}$, and $\langle x, x\rangle > 0$ for $x \ne 0$, is an **inner product space**; $\|x\| = \langle x, x\rangle^{1/2}$.

Theorem 6.17 (6.5.1, Cauchy–Schwarz)

$|\langle x, y\rangle| \le \|x\|\|y\|$ in a complex inner product space; hence $\|x + y\| \le \|x\| + \|y\|$.

Proof. If $y \ne 0$, put $\lambda = \langle x, y\rangle / \|y\|^2$; then $0 \le \|x - \lambda y\|^2 = \|x\|^2 - |\langle x, y\rangle|^2/\|y\|^2$. Then $\|x+y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Summation by parts: with $F_n = \sum_{k\le n} f_k$ ($F_0 = 0$),

$$\sum_{k=m}^{n} f_k g_k = F_n g_n - F_{m-1} g_m + \sum_{k=m}^{n-1} F_k (g_k - g_{k+1}). \tag{†}$$

Theorem 6.18 (6.5.2, Dirichlet–Dedekind test)

Let $f_n, g_n : X \to \mathbb{C}$ satisfy:
(a) the partial sums F_n of $\sum f_n$ are uniformly bounded, $|F_n(x)| \le K$;
(b) $g_n \to 0$ uniformly;
(c) $\sum_n |g_n(x) - g_{n+1}(x)|$ converges uniformly on X. Then $\sum f_n g_n$ converges uniformly on X.

Proof. By (†), $|\sum_{k=m}^{n} f_k g_k| \le K|g_n| + K|g_m| + K\sum_{k=m}^{n-1}|g_k - g_{k+1}|$, which is uniformly small for large m; apply Theorem 6.2. ■

Corollary 6.19 (6.5.3)

If the partial sums of $\sum a_n$ are bounded and b_n decreases to 0, then $\sum a_n b_n$ converges.

Proof. Here $\sum |b_n - b_{n+1}| = b_1$; apply Theorem 6.18 with constant functions. ■

This contains the alternating series test (Theorem 2.17 of Volume I). For $x \notin 2\pi\mathbb{Z}$,

$$\sum_{k=1}^{n} e^{ikx} = e^{ix}\frac{e^{inx} - 1}{e^{ix} - 1}, \qquad \Big|\sum_{k=1}^{n} e^{ikx}\Big| \le \frac{1}{|\sin(x/2)|}, \tag{‡}$$

so the partial sums of $\sum \sin kx$ and $\sum \cos kx$ are bounded by $1/\sin(\delta/2)$ on $[\delta, 2\pi - \delta]$.

Theorem 6.20 (6.5.4)

If $a_n \to 0$ and $\sum |a_n - a_{n-1}| < \infty$, then for every $\delta \in (0, \pi)$ the series $\sum a_n \sin nx$ and $\sum a_n \cos nx$ converge uniformly on $[\delta, 2\pi - \delta]$.

Proof. Theorem 6.18 with $f_n(x) = e^{inx}$ (bounded partial sums by (‡)) and constant $g_n = a_n$; take real and imaginary parts. ■

Theorem 6.21 (6.5.5)

Let a_n decrease to 0. Then $\sum a_n \sin nx$ converges uniformly on $\mathbb{R}$ iff $na_n \to 0$.

Proof. ($\Rightarrow$) Given ϵ, uniform convergence gives $|\sum_{n/2 \le k \le n} a_k \sin kx| < \epsilon$ for all x and all large n. Take $x = \pi/(2n)$: for $n/2 \le k \le n$, $kx \in [\pi/4, \pi/2]$, so $\sin kx \ge 1/\sqrt{2}$, and $a_k \ge a_n$. There are at least $n/2$ such k, so $\frac{n}{2} a_n/\sqrt{2} < \epsilon$. Hence $na_n \to 0$.

($\Leftarrow$) Since a_k decreases, (†) with $f_k = \sin kx$ and (‡) give $|\sum_{k=m}^{n} a_k \sin kx| \le 2a_m/\sin(x/2) \le 2\pi a_m/x$ for $0 < x \le \pi$ (using $\sin(x/2) \ge x/\pi$). Let $\epsilon_m = \sup_{k \ge m} k a_k \to 0$. By oddness and periodicity it suffices to bound $R_m(x) = \sum_{k \ge m} a_k \sin kx$ for $x \in (0, \pi]$; let N be the largest integer $\le \pi/x$. If $m > N$, then $m \ge N + 1 > \pi/x$, so $|R_m(x)| \le 2\pi a_m/x < 2m a_m \le 2\epsilon_m$. If $m \le N$, split $R_m = \sum_{k=m}^{N} + \sum_{k>N}$. In the first sum $|a_k \sin kx| \le k a_k x \le \epsilon_m x$, and there are at most $N \le \pi/x$ terms, so it is at most $\pi\epsilon_m$; the second is at most $2\pi a_{N+1}/x < 2(N+1)a_{N+1} \le 2\epsilon_m$. So $\sup_x |R_m(x)| \le (\pi + 2)\epsilon_m \to 0$. ■

Theorem 6.22 (6.5.6, Abel's test)

Let $f_n : X \to \mathbb{C}$ and $g_n : X \to \mathbb{R}$ satisfy:
(a) $\sum f_n$ converges uniformly;
(b) $g_{n+1}(x) \le g_n(x)$ for all x, n;
(c) (g_n) is uniformly bounded, $|g_n| \le M$. Then $\sum f_n g_n$ converges uniformly.

Proof. Apply (†) to the tails: with $R_k = \sum_{j=m}^{k} f_j$ ($R_{m-1} = 0$), $\sum_{k=m}^{n} f_k g_k = R_n g_n + \sum_{k=m}^{n-1} R_k(g_k - g_{k+1})$. Given ϵ, for m large $|R_k(x)| < \epsilon$ for all $k \geq m$, x, by (a). Since $g_k - g_{k+1} \geq 0$, the sum is at most $\epsilon M + \epsilon(g_m - g_n) \leq 3M\epsilon$ in absolute value. ■

6.6 Selected exercises

Motivation. These problems from the textbook's exercise set practise the main tools of the chapter: compactness upgrades monotone pointwise convergence to uniform convergence, termwise integration locates zeros, Abel's theorem evaluates a Cauchy product that converges only conditionally, and a growth condition on the coefficients gives a converse of Abel's theorem.

Exercise 6.6.6 (Dini's theorem)

Let $K \subset \mathbb{R}^n$ be compact and let $f_n : K \to [0, \infty)$ be continuous with $f_{n+1} \leq f_n$ for all n. Show that if $f_n \to 0$ pointwise on K, then $f_n \to 0$ uniformly on K.

Solution.

Step 1. Fix $\epsilon > 0$ and let $U_n = \{x \in K : f_n(x) < \epsilon\}$. By continuity, $f_n^{-1}((-\infty, \epsilon))$ is relatively open in K, so $U_n = V_n \cap K$ for some open $V_n \subset \mathbb{R}^n$. Since $f_{n+1} \leq f_n$, $U_n \subset U_{n+1}$. Since $f_n(x) \to 0$ for every x, every $x \in K$ lies in some U_n, so $K \subset \bigcup_n V_n$.

Step 2. By compactness, $K \subset V_{n_1} \cup \cdots \cup V_{n_r}$. With $N = \max_j n_j$ and the monotonicity of the U_n, $K = U_{n_1} \cup \cdots \cup U_{n_r} \subset U_N$.

Step 3. For $n \geq N$ and $x \in K$, $0 \leq f_n(x) \leq f_N(x) < \epsilon$. Hence $\|f_n\|_{\sup} \leq \epsilon$ for $n \geq N$, i.e. $f_n \to 0$ uniformly.

No hypothesis can be dropped: x^n on $[0, 1)$ (not compact) and $nx(1-x)^n$ on $[0, 1]$ (not monotone in n; its maximum tends to e^{-1}) tend to 0 pointwise but not uniformly. ■

Exercise 6.6.12 (A zero from an integral)

Let $f(x) = \sum_{n \geq 0} a_n x^n$ have radius of convergence greater than 1, with real a_n satisfying

$$0 < a_0 < -\sum_{k=1}^{\infty} \frac{a_{2k}}{2k+1}.$$

Show that f has at least one zero in $(-1, 1)$.

Solution.

Step 1. Since 1 is less than the radius, the series converges uniformly on $[-1, 1]$ (Theorem 6.12), and in particular $\sum_k |a_{2k}| < \infty$, so the series in the hypothesis converges. By Proposition 6.5 we may integrate termwise. As $\int_{-1}^{1} x^n dx$ is 0 for odd n and $\frac{2}{n+1}$ for even n,

$$\int_{-1}^{1} f(x)\, dx = 2\Big(a_0 + \sum_{k=1}^{\infty} \frac{a_{2k}}{2k+1}\Big) < 0.$$

Step 2. Suppose f had no zero in $(-1, 1)$. Since f is continuous and $f(0) = a_0 > 0$, the intermediate value theorem gives $f > 0$ on $(-1, 1)$, hence $\int_{-1}^{1} f \geq 0$, contradicting Step 1. So $f(c) = 0$ for some $c \in (-1, 1)$. ■

Exercise 6.6.13 (Square of the alternating harmonic series)

Let $a_n = \frac{(-1)^{n+1}}{n+1}$ $(n \geq 0)$. Show that the Cauchy product of $\sum a_n$ with itself is

$$2\sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m+1}\Big(1+\frac{1}{2}+\cdots+\frac{1}{m}\Big),$$

that this series converges, and that its sum is $(\log 2)^2$.

Solution.

Step 1. (The product.) Write $H_m = 1+\frac{1}{2}+\cdots+\frac{1}{m}$. For $n \geq 0$,

$$c_n = \sum_{k=0}^{n} a_k a_{n-k} = (-1)^n \sum_{k=0}^{n} \frac{1}{(k+1)(n+1-k)}.$$

Using $\frac{1}{(k+1)(n+1-k)} = \frac{1}{n+2}\big(\frac{1}{k+1}+\frac{1}{n+1-k}\big)$, the last sum equals $\frac{2H_{n+1}}{n+2}$. With $m = n+1$, $c_{m-1} = 2(-1)^{m+1}\frac{H_m}{m+1}$, which is the stated series.

Step 2. (Convergence.) Let $b_m = \frac{H_m}{m+1}$. Since $H_{m+1} = H_m + \frac{1}{m+1}$,

$$b_m - b_{m+1} = \frac{(m+2)H_m - (m+1)H_{m+1}}{(m+1)(m+2)} = \frac{H_m - 1}{(m+1)(m+2)} \geq 0.$$

Also $\frac{1}{k} \leq \int_{k-1}^{k} \frac{dx}{x}$ for $k \geq 2$ gives $H_m \leq 1+\log m$, so $b_m \to 0$. By the alternating series test (Corollary 6.19), $\sum(-1)^{m+1}b_m$ converges.

Step 3. (The sum.) By Example 6.4, $\sum a_n = -\log 2$. The series $\sum a_n$ converges only conditionally, so Mertens' theorem (Proposition 6.14) does not apply; but all three series converge, so Corollary 6.16 gives the sum $(-\log 2)^2 = (\log 2)^2$. ∎

Exercise 6.6.14 (Tauber's theorem)

Let $f(x) = \sum_{n\geq 0} a_n x^n$ converge on $(-1,1)$, and suppose $\lim_{x\to 1^-} f(x) = L$ and $\lim_n na_n = 0$. Let $s_n = \sum_{k\leq n} a_k$.
(a) Show that $\lim_{n\to\infty}(s_n - f(1-\frac{1}{n})) = 0$.
(b) Show that $\sum_{n\geq 0} a_n = L$.

Solution.

(a) Let $x_n = 1-\frac{1}{n}$ $(n \geq 2)$. Then

$$s_n - f(x_n) = \sum_{k\leq n} a_k(1-x_n^k) - \sum_{k>n} a_k x_n^k.$$

Step 1. For $0 \leq x < 1$, $1-x^k = (1-x)(1+x+\cdots+x^{k-1}) \leq k(1-x)$. Hence

$$\Big|\sum_{k\leq n} a_k(1-x_n^k)\Big| \leq \frac{1}{n}\sum_{k=0}^{n} k|a_k| = \frac{n+1}{n}\cdot\frac{1}{n+1}\sum_{k=0}^{n} k|a_k| \to 0,$$

because $k|a_k| \to 0$ and arithmetic means of a null sequence tend to 0 (Lemma 9.12).

Step 2. Let $\epsilon_n = \sup_{k>n} k|a_k|$, so $\epsilon_n \to 0$. For $k > n$, $|a_k| \leq \epsilon_n/k < \epsilon_n/n$, so

$$\Big|\sum_{k>n} a_k x_n^k\Big| \leq \frac{\epsilon_n}{n}\sum_{k\geq 0} x_n^k = \frac{\epsilon_n}{n}\cdot\frac{1}{1-x_n} = \epsilon_n \to 0.$$

Steps 1 and 2 give (a).

(b) $f(x_n) \to L$ since $x_n \to 1^-$, so by (a), $s_n = f(x_n) + (s_n - f(x_n)) \to L$.

Without a condition such as $na_n \to 0$ the converse of Abel's theorem fails: $a_n = (-1)^n$ gives $f(x) = \frac{1}{1+x} \to \frac{1}{2}$, while $\sum a_n$ diverges. ■

Chapter 7. Function Spaces

7.1 The space $C(X)$

Motivation. Thinking of functions as points of a space with a metric gives a geometric language for approximation and convergence. The space $C(X)$ with the sup norm is complete, and the Stone–Weierstrass theorem says polynomials are dense in it.

A **norm** on a real or complex vector space V is a function $\|\cdot\|$ with $\|v\| \geq 0$ (equality iff $v = 0$), $\|\lambda v\| = |\lambda|\|v\|$, and $\|u + v\| \leq \|u\| + \|v\|$; V is then a **normed space**, with distance $\|u - v\|$, and convergence, Cauchy sequences, open and closed sets are defined as in $\mathbb{R}^n$. A normed space in which every Cauchy sequence converges is **complete** (a **Banach space**). For compact $X \subset \mathbb{R}^n$, $C(X)$ is the space of continuous $f : X \to \mathbb{C}$ with $\|f\|_{\sup} = \max_X |f|$; convergence in this norm is uniform convergence.

Theorem 7.1 (7.1.1)

For compact X, every Cauchy sequence in $C(X)$ converges: $C(X)$ is complete.

Proof. A Cauchy sequence in $C(X)$ is uniformly Cauchy, so converges uniformly (Theorem 6.2) to a function that is continuous (Theorem 6.1). ■

The Heine–Borel theorem fails in $C(X)$: the functions $f_n(x) = x^n$ on $[0, 1]$ form a closed bounded set (no subsequence converges uniformly, since the pointwise limit is discontinuous). The missing ingredient is equicontinuity. A family $\mathcal{F} \subset C(X)$ is **equicontinuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \epsilon$ for all $f \in \mathcal{F}$ and $\|x - y\| < \delta$; it is **bounded** if $\sup_{f\in\mathcal{F}} \|f\|_{\sup} < \infty$.

Proposition 7.2 (7.1.2)

If X is compact and (f_n) converges uniformly in $C(X)$, then $\{f_n\}$ is equicontinuous.

Proof. Given ϵ, choose N with $\|f_n - f_N\|_{\sup} < \epsilon/3$ for $n \geq N$ (Theorem 6.2). The finitely many functions $f_1, \dots, f_N$ are uniformly continuous (Theorem 3.18 of Volume I), so some δ works for all of them with $\epsilon/3$. For $n > N$ and $\|x - y\| < \delta$, $|f_n(x) - f_n(y)| \leq |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < \epsilon$. ■

Theorem 7.3 (7.1.3, Arzelà–Ascoli)

If X is compact and $\mathcal{F} \subset C(X)$ is bounded and equicontinuous, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.

Proof. Let (f_n) be in $\mathcal{F}$, $|f_n| \leq M$. X has a countable dense subset $E = \{p_1, p_2, \dots\}$ (for each k, finitely many balls of radius $1/k$ cover X; take their centres in X). The bounded sequence $(f_n(p_1))$ has a convergent subsequence $(f_{1,n})$; $(f_{1,n}(p_2))$ has a convergent subsequence $(f_{2,n})$; and so on. The diagonal sequence $g_n = f_{n,n}$ is, from the k-th term on, a subsequence of $(f_{k,n})$, so $(g_n(p_k))$ converges for every k. Given ϵ, take δ from equicontinuity for $\epsilon/3$, and finitely many $p_{k_1}, \dots, p_{k_r}$ with every $x \in X$ within δ of one of them. Choose N with $|g_m(p_{k_j}) - g_n(p_{k_j})| < \epsilon/3$ for $m, n \geq N$ and all j. For any x, pick $p = p_{k_j}$ with $\|x - p\| < \delta$; then $|g_m(x) - g_n(x)| \leq |g_m(x) - g_m(p)| + |g_m(p) - g_n(p)| + |g_n(p) - g_n(x)| < \epsilon$. So (g_n) is uniformly Cauchy and converges in $C(X)$ (Theorem 7.1). ■

Combined with Proposition 7.2: a subset of $C(X)$ has compact closure iff it is bounded and equicontinuous.

Theorem 7.4 (7.1.4, Weierstrass approximation)

For every continuous $f : [a, b] \to \mathbb{R}$ there are polynomials $P_n \to f$ uniformly on $[a, b]$.

Proof. By an affine change of variable take $[a, b] = [0, 1]$, and subtracting the linear function $f(0) + x(f(1) - f(0))$ assume $f(0) = f(1) = 0$. Extend f by 0 outside $[0, 1]$; it is uniformly continuous on $\mathbb{R}$. Let $Q_n(x) = c_n(1 - x^2)^n$ on $[-1, 1]$ with c_n chosen so that $\int_{-1}^{1} Q_n = 1$. Since $(1 - x^2)^n \geq 1 - nx^2$,

$$\int_{-1}^{1} (1 - x^2)^n dx \geq 2 \int_{0}^{1/\sqrt{n}} (1 - nx^2) dx = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}},$$

so $c_n < \sqrt{n}$, and for $\delta \leq |x| \leq 1$, $Q_n(x) \leq \sqrt{n}(1 - \delta^2)^n \to 0$ uniformly. Let

$$P_n(x) = \int_{-1}^{1} f(x + t) Q_n(t)\, dt = \int_{0}^{1} f(t) Q_n(t - x)\, dt \qquad (0 \leq x \leq 1),$$

a polynomial in x. Given ϵ, choose δ with $|f(y) - f(x)| < \epsilon/2$ for $|y - x| < \delta$, and let $M = \max|f|$. Then

$$|P_n(x) - f(x)| \leq \int_{-1}^{1} |f(x + t) - f(x)| Q_n(t)\, dt \leq 2M \cdot 2\sqrt{n}(1 - \delta^2)^n + \frac{\epsilon}{2} < \epsilon$$

for large n, uniformly in $x \in [0, 1]$. ■

For complex-valued f, approximate real and imaginary parts. Consequently $C[a, b]$ has a countable dense set (polynomials with rational coefficients).

7.2 Sequence spaces

Motivation. The sequence spaces ℓ^p are the simplest infinite-dimensional normed spaces. Hölder's and Minkowski's inequalities make them normed and complete, and they are models for the function spaces of Chapter 10.

$\ell^1 = \ell^1(\mathbb{N})$ is the space of sequences $a = (a(n))$ with $\|a\|_1 = \sum|a(n)| < \infty$, and ℓ^2 those with $\|a\|_2 = (\sum|a(n)|^2)^{1/2} < \infty$. ℓ^2 is a vector space with inner product $\langle a, b\rangle = \sum a(n)\overline{b(n)}$, which converges absolutely since $|a(n)b(n)| \leq \frac{1}{2}(|a(n)|^2 + |b(n)|^2)$; $\|\cdot\|_2$ is the associated norm (Theorem 6.17).

Theorem 7.5 (7.2.1)

ℓ^1 is complete.

Proof. Let (a_m) be Cauchy. Since $|a_m(n) - a_k(n)| \leq \|a_m - a_k\|_1$, each $(a_m(n))_m$ converges to some $a(n)$. Given ϵ, choose N with $\|a_m - a_k\|_1 < \epsilon$ for $m, k \geq N$. For each L, $\sum_{n \leq L}|a_m(n) - a_k(n)| < \epsilon$; let $k \to \infty$ and then $L \to \infty$: $\sum_n|a_m(n) - a(n)| \leq \epsilon$ for $m \geq N$. So $a = a_m - (a_m - a) \in \ell^1$ and $a_m \to a$. ■

Theorem 7.6 (7.2.2)

ℓ^2 is complete.

Proof. As in Theorem 7.5: the coordinates converge, and $\sum_{n \leq L}|a_m(n) - a_k(n)|^2 < \epsilon^2$ passes to the limit in k and then in L. ■

The same proof shows that ℓ^∞ (bounded sequences with the sup norm) is complete. The unit vectors e_n in ℓ^2 form a bounded sequence with $\|e_m - e_n\|_2 = \sqrt{2}$, so without convergent subsequence: closed bounded sets in infinite-dimensional spaces need not be compact.

7.3 Improper integrals

Motivation. Many important integrals, such as $\int_0^\infty e^{-x^2}dx$, run over infinite intervals or involve unbounded integrands. Improper integrals extend Riemann integration to them, with convergence tests parallel to those for series.

Let I be an interval and $f : I \to \mathbb{C}$ Riemann integrable on every compact subinterval. If $I = [a, b)$ ($b \le \infty$), the **improper integral** $\int_I f = \lim_{c\to b^-} \int_a^c f$ when the limit exists; similarly at a left end point, and for open intervals both ends are treated separately. We write $\int_I |f| < \infty$ when this improper integral converges (**absolute convergence**).

Proposition 7.7 (7.3.1)

If $0 \le f \le g$ on I and $\int_I g$ converges, then $\int_I f$ converges.

Proof. $c \mapsto \int_a^c f$ is increasing and bounded by $\int_I g$, so it has a limit as $c \to b^-$ (as for monotone sequences). ∎

Proposition 7.8 (7.3.2, comparison test)

If $|f| \le g$ and $\int_I g$ converges, then $\int_I |f|$ and $\int_I f$ converge and $|\int_I f| \le \int_I |f| \le \int_I g$.

Proof. $\int_I |f|$ converges by Proposition 7.7. Cauchy criterion: for $a < c < c' < b$, $|\int_c^{c'} f| \le \int_c^{c'} |f| \to 0$ as $c, c' \to b$, so $\lim \int_a^c f$ exists (as in Theorem 2.14 of Volume I). The inequality passes to the limit from finite intervals (Theorem 5.5 of Volume I). ∎

Example 7.1

$\int_1^\infty x^{-p}dx < \infty$ iff $p > 1$, and $\int_0^1 x^{-p}dx < \infty$ iff $p < 1$. $\int_0^\infty \frac{\sin x}{x}\,dx$ converges (integrate by parts: $\int_1^c \frac{\sin x}{x}dx = \frac{\cos 1}{1} - \frac{\cos c}{c} - \int_1^c \frac{\cos x}{x^2}dx$) but not absolutely (on $[k\pi, (k+1)\pi]$, $\int |\sin x|/x \ge \frac{2}{(k+1)\pi}$).

Proposition 7.9 (7.3.3, integral test)

If $f : [1, \infty) \to (0, \infty)$ is decreasing, then $\sum f(n)$ and $\int_1^\infty f$ converge or diverge together; indeed $\sum_{n=2}^N f(n) \le \int_1^N f \le \sum_{n=1}^{N-1} f(n)$, and if $f \to 0$, then $\sum_{n\le N} f(n) - \int_1^N f$ converges.

Proof. On $[n, n+1]$, $f(n+1) \le f \le f(n)$; integrate and sum. The differences $d_N = \sum_{n\le N} f(n) - \int_1^N f$ decrease ($d_N - d_{N+1} = \int_N^{N+1} f - f(N+1) \ge 0$) and are $\ge f(N) > 0$, so converge. ∎

For $f(x) = 1/x$ this gives Euler's constant $\gamma = \lim(1 + \frac{1}{2} + \cdots + \frac{1}{N} - \log N)$.

Theorem 7.10 (7.3.4)

If $\int_I |f| < \infty$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $\int_x^y |f| < \epsilon$ whenever $x < y$ in I and $y - x < \delta$.

Proof. Choose a compact $[c, d] \subset I$ with $\int_{I-[c,d]} |f| < \epsilon/2$; f is bounded on $[c, d]$, say by K. With $\delta = \epsilon/(2K + 1)$, $\int_x^y |f| \le \int_{[x,y]\cap[c,d]} |f| + \int_{I-[c,d]} |f| < K\delta + \epsilon/2 < \epsilon$. ∎

7.4 Spaces of integrable functions

Motivation. Just as ℓ^p measures sequences, the integral defines norms on functions. These spaces are not complete with the Riemann integral, a defect that motivates the Lebesgue integral of Chapter 10.

Proposition 7.11 (7.4.1)

For a normed space X the following are equivalent:
(a) every Cauchy sequence in X converges;
(b) whenever $\sum\|x_n\| < \infty$, the series $\sum x_n$ converges in X.

Proof. (a)⇒(b): the partial sums are Cauchy, since $\|\sum_{m<k\le n} x_k\| \le \sum_{m<k\le n}\|x_k\|$.

(b)⇒(a): given a Cauchy (y_n), choose $n_1 < n_2 < \cdots$ with $\|y_m - y_n\| < 2^{-k}$ for $m, n \ge n_k$. With $x_1 = y_{n_1}$, $x_{k+1} = y_{n_{k+1}} - y_{n_k}$, $\sum\|x_k\| < \infty$, so $y_{n_k} = \sum_{j\le k} x_j$ converges; a Cauchy sequence with a convergent subsequence converges (Lemma 2.13 of Volume I, same proof). ∎

Let $\mathcal{R}^1(I)$ and $\mathcal{R}^2(I)$ be the locally Riemann integrable $f : I \to \mathbb{C}$ with $\int_I|f| < \infty$, resp. $\int_I|f|^2 < \infty$, with $\|f\|_1 = \int_I|f|$ and $\|f\|_2 = (\int_I|f|^2)^{1/2}$. These are seminorms ($\|f\| = 0$ does not force $f = 0$), and $\mathcal{R}^2$ carries the inner product $\langle f, g\rangle = \int_I f\bar{g}$. Neither space is complete; this is the reason for introducing the Lebesgue integral (Chapter 10). A **step function** is a finite linear combination of characteristic functions of bounded intervals.

Lemma 7.12 (7.4.2)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a step function s with $\|f - s\|_2 < \epsilon$.

Proof. Treating real and imaginary parts separately, let f be real. Choose a compact $[a, b] \subset I$ with $\int_{I-[a,b]}|f|^2 < \epsilon^2/4$. On $[a, b]$, f is Riemann integrable, so bounded by some K; choose a partition with $U(f, P) - L(f, P) < \epsilon^2/(8K)$ and let $s = m_i$ on the i-th interval (and 0 off $[a, b]$). Then $0 \le f - s \le 2K$ on $[a, b]$, so $\int_a^b|f - s|^2 \le 2K\int_a^b(f - s) \le 2K(U - L) < \epsilon^2/4$. Altogether $\|f - s\|_2^2 < \epsilon^2/2$. ∎

Lemma 7.13 (7.4.3)

For a step function s on I and $\epsilon > 0$ there is a continuous g with $\|s - g\|_2 < \epsilon$ (and $g = 0$ outside a compact subinterval).

Proof. It suffices to treat $s = \chi_{[a,b]}$ (with any type of end points). Let g be 1 on $[a + \eta, b - \eta]$, 0 outside (a, b), and linear in between; then $|s - g| \le 1$ and differs from 0 only on two intervals of length η, so $\|s - g\|_2^2 \le 2\eta$. ∎

Theorem 7.14 (7.4.4)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a continuous g with $\|f - g\|_2 < \epsilon$.

Proof. Combine Lemmas 7.12 and 7.13 with the triangle inequality for $\|\cdot\|_2$. ∎

Corollary 7.15 (7.4.5)

For $f \in \mathcal{R}^2[a, b]$ there are polynomials P_n with $\|f - P_n\|_2 \to 0$.

Proof. Choose continuous g with $\|f-g\|_2<\epsilon$ and a polynomial P with $\|g-P\|_{\sup}<\epsilon$ (Theorem 7.4); then $\|g-P\|_2\le\epsilon\sqrt{b-a}$. ∎

Theorem 7.16 (7.4.6)

For $f\in\mathcal{R}^1(I)$ and $\epsilon>0$ there is a step function s with $\|f-s\|_1<\epsilon$, and a continuous g vanishing outside a compact subinterval with $\|f-g\|_1<\epsilon$. For $f\in\mathcal{R}^1[a,b]$ there are polynomials P with $\|f-P\|_1<\epsilon$.

Proof. As in Lemma 7.12, with $\int_a^b|f-s|=\int_a^b(f-s)\le U-L$ for the lower step function. For the continuous approximation of $\chi_{[a,b]}$ the error in $\|\cdot\|_1$ is at most η. Polynomials: $\|g-P\|_1\le(b-a)\|g-P\|_{\sup}$. ∎

7.5 Selected exercises

Motivation. These problems from the textbook's exercise set use the Weierstrass theorem and the geometry of function and sequence spaces: moments determine a continuous function, Bernstein polynomials give a constructive proof of the Weierstrass theorem, and compactness in ℓ^2 depends on how fast the coordinates are forced to decay.

Exercise 7.5.1 (Moments determine a function)

Let $f:[0,1]\to\mathbb{R}$ be continuous with $\int_0^1 f(x)x^n\,dx=0$ for $n=0,1,2,\dots$. Show that $f=0$.

Solution.

Step 1. By linearity, $\int_0^1 fP=0$ for every polynomial P.

Step 2. By Theorem 7.4 there are polynomials $P_n\to f$ uniformly on $[0,1]$. Then

$$\int_0^1 f^2=\int_0^1 f(f-P_n)\le\|f\|_{\sup}\|f-P_n\|_{\sup}\to 0,$$

so $\int_0^1 f^2=0$.

Step 3. If $f(x_0)\neq 0$, then by continuity $f^2>\tfrac12 f(x_0)^2$ on some interval $J\subset[0,1]$ of length $\eta>0$, and $\int_0^1 f^2\ge\tfrac12 f(x_0)^2\eta>0$, a contradiction. Hence $f=0$. ∎

Exercise 7.5.5 (Weights concentrating at an end point)

Let $f:[0,1]\to\mathbb{R}$ be continuous. Show that

$$\lim_{n\to\infty}\frac{\int_0^1 x^n f(x)\,dx}{\int_0^1 x^n\,dx}=f(1).$$

Solution.

Step 1. Since $\int_0^1 x^n dx=\frac{1}{n+1}$, the quotient minus $f(1)$ equals $(n+1)\int_0^1 x^n(f(x)-f(1))dx$.

Step 2. Let $M=\max|f|$ and $\epsilon>0$. By continuity at 1 choose $\delta\in(0,1)$ with $|f(x)-f(1)|<\epsilon$ for $1-\delta\le x\le 1$. Then

$$(n+1)\int_0^{1-\delta}x^n|f(x)-f(1)|\,dx\le 2M(1-\delta)^{n+1},$$

$$(n+1)\int_{1-\delta}^{1} x^n |f(x)-f(1)|\,dx \le \epsilon\,(n+1)\int_0^1 x^n dx = \epsilon.$$

Step 3. The first bound tends to 0, so the absolute value in Step 1 is less than 2ϵ for large n. As ϵ is arbitrary, the limit is $f(1)$. ■

Exercise 7.5.7 (Bernstein polynomials)

Let $f:[0,1]\to\mathbb{R}$ be continuous, $r_k(x)=\binom{n}{k}x^k(1-x)^{n-k}$, and $p_n(x)=\sum_{k=0}^n \binom{n}{k} f\left(\frac{k}{n}\right)x^k(1-x)^{n-k}$.
(a) Show that $\sum_{k=0}^n (k-nx)^2 r_k(x) = nx(1-x)$.
(b) Show that $p_n \to f$ uniformly on $[0,1]$.

Solution.

(a)

Step 1. By the binomial theorem, $\sum_k r_k(x) = (x+(1-x))^n = 1$.

Step 2. From $k\binom{n}{k} = n\binom{n-1}{k-1}$ and $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$, the binomial theorem again gives

$$\sum_k k\,r_k(x) = nx, \qquad \sum_k k(k-1)\,r_k(x) = n(n-1)x^2.$$

Step 3. Expanding $(k-nx)^2 = k(k-1)+k-2nxk+n^2x^2$,

$$\sum_k (k-nx)^2 r_k(x) = n(n-1)x^2 + nx - 2n^2x^2 + n^2x^2 = nx(1-x).$$

(b)

Step 1. By (a), Step 1, $f(x)-p_n(x) = \sum_k \left(f(x)-f\left(\frac{k}{n}\right)\right) r_k(x)$. Let $M=\max|f|$ and $\epsilon>0$; by uniform continuity choose $\delta>0$ with $|f(x)-f(y)|<\epsilon$ whenever $|x-y|<\delta$.

Step 2. The indices with $\left|\frac{k}{n}-x\right|<\delta$ contribute at most $\epsilon\sum_k r_k(x) = \epsilon$.

Step 3. For the other indices $(k-nx)^2 \ge n^2\delta^2$, so by (a) their contribution is at most

$$2M\sum_k \frac{(k-nx)^2}{n^2\delta^2} r_k(x) = \frac{2Mx(1-x)}{n\delta^2} \le \frac{M}{2n\delta^2}.$$

Step 4. Hence $\|f-p_n\|_{\sup} \le \epsilon + \frac{M}{2n\delta^2} < 2\epsilon$ for large n. This is a constructive proof of Theorem 7.4. ■

Exercise 7.5.9 (Compact and non-compact boxes in ℓ^2)

Show that $K=\{a\in\ell^2 : |a(n)|\le \frac{1}{n},\ n=1,2,\dots\}$ is compact, i.e. every sequence in K has a subsequence converging in ℓ^2 to a point of K. Is $L=\{a\in\ell^2 : |a(n)|\le \frac{1}{\sqrt{n}},\ n=1,2,\dots\}$ compact?

Solution.

(a) K is compact.

Step 1. Let (a_m) be a sequence in K. Each coordinate sequence $(a_m(n))_m$ is bounded by $\frac{1}{n}$. By the diagonal argument of Theorem 7.3, there is a subsequence (b_j) with $b_j(n)\to a(n)$ for every n.

Step 2. $|a(n)|\le\frac{1}{n}$, so $\sum|a(n)|^2 \le \sum n^{-2} < \infty$ and $a\in K$.

Step 3. Given $\epsilon > 0$ choose N with $\sum_{n>N} \frac{4}{n^2} < \frac{\epsilon^2}{2}$, then j_0 with $\sum_{n\le N}|b_j(n)-a(n)|^2 < \frac{\epsilon^2}{2}$ for $j \ge j_0$. Since $|b_j(n)-a(n)| \le \frac{2}{n}$,

$$\|b_j - a\|_2^2 = \sum_{n\le N} |b_j(n)-a(n)|^2 + \sum_{n>N} |b_j(n)-a(n)|^2 < \epsilon^2 \qquad (j \ge j_0).$$

(b) L is not compact.

Step 1. For $m \ge 1$ let $c_m(n) = \frac{1}{\sqrt{n}}$ for $2^m < n \le 2^{m+1}$ and $c_m(n) = 0$ otherwise. Then $c_m \in L$ and

$$\|c_m\|_2^2 = \sum_{2^m < n \le 2^{m+1}} \frac{1}{n} \ge 2^m \cdot \frac{1}{2^{m+1}} = \frac{1}{2}.$$

Step 2. The c_m have disjoint supports, so $\|c_m - c_{m'}\|_2^2 = \|c_m\|_2^2 + \|c_{m'}\|_2^2 \ge 1$ for $m \ne m'$. No subsequence is Cauchy, so none converges. ∎

Chapter 8. Functions Defined by Integrals

8.1 Integrals depending on a parameter

Motivation. Functions like $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}dt$ are defined by integrals with a parameter. We need conditions under which such functions are continuous and can be differentiated under the integral sign.

For f on $[a,b]\times[c,d]$ let $\phi(x) = \int_c^d f(x,t)\,dt$. Write $D_1 f = \partial f/\partial x$.

Proposition 8.1 (8.1.1)

Let $f : [a,b]\times[c,d] \to \mathbb{R}$ be continuous.
(a) ϕ is continuous on $[a,b]$.
(b) If $D_1 f$ exists and is continuous on $(a,b)\times[c,d]$, then ϕ is differentiable on (a,b) and $\phi'(x) = \int_c^d D_1 f(x,t)\,dt$.

Proof. (a) f is uniformly continuous on the compact rectangle (Theorem 3.18 of Volume I): given ϵ there is δ with $|f(x,t) - f(x',t)| < \epsilon$ for $|x-x'| < \delta$ and all t; then $|\phi(x) - \phi(x')| \le \epsilon(d-c)$.

(b) Fix x and a compact $[x-\eta, x+\eta] \subset (a,b)$, on which $D_1 f$ is uniformly continuous. For $0 < |h| < \eta$, the mean value theorem gives $\frac{f(x+h,t)-f(x,t)}{h} = D_1 f(x+\theta h, t)$ with $0 < \theta = \theta(h,t) < 1$, so

$$\Big|\frac{\phi(x+h)-\phi(x)}{h} - \int_c^d D_1 f(x,t)\,dt\Big| \le \int_c^d |D_1 f(x+\theta h,t) - D_1 f(x,t)|\,dt \le \epsilon(d-c)$$

once $|h|$ is small. ∎

Example 8.1

Differentiating $\int_0^1 t^x\,dt = \frac{1}{x+1}$ n times in $x > -1$ gives $\int_0^1 t^x(\log t)^n dt = (-1)^n n!/(x+1)^{n+1}$. Interchanging the order, $\int_a^b \phi(x)\,dx = \int_c^d \int_a^b f(x,t)\,dx\,dt$ for continuous f: both sides, as functions of the upper limit b, vanish at $b = a$ and have the same derivative $\int_c^d f(b,t)\,dt$ (by (a), (b) and Theorem 5.11 of Volume I).

For f continuous on $[a,b]\times[c,d)$ $(d \le \infty)$, consider the improper integral $\phi(x) = \int_c^d f(x,t)\,dt$.

Theorem 8.2 (8.1.2)

Let $f : [a,b]\times[c,d) \to \mathbb{R}$ be continuous, and suppose there is $g : [c,d) \to \mathbb{R}$ with (a) $|f(x,t)| \le g(t)$ for all (x,t) and (b) $\int_c^d g < \infty$. Then $\int_c^d f(x,t)\,dt$ converges for each x and ϕ is continuous on $[a,b]$.

Proof. Convergence follows from the comparison test (Proposition 7.8). Given ϵ, choose $d' < d$ with $\int_{d'}^d g < \epsilon$. Then $|\phi(x) - \phi_{d'}(x)| < \epsilon$ for all x, where $\phi_{d'}(x) = \int_c^{d'} f(x,t)\,dt$ is continuous by Proposition 8.1. So ϕ is a uniform limit of continuous functions (Theorem 6.1). ∎

Theorem 8.3 (8.1.3)

Let $f : [a,b]\times[c,d) \to \mathbb{R}$ be continuous, $D_1 f$ exist and be continuous on $(a,b)\times[c,d)$, and $\int_c^d f(x_0,t)\,dt$ converge for some $x_0 \in (a,b)$. If there is $h : [c,d) \to \mathbb{R}$ with (a) $|D_1 f(x,t)| \le h(t)$ and (b) $\int_c^d h < \infty$, then $\int_c^d f(x,t)\,dt$ converges for each $x \in (a,b)$, and ϕ is differentiable with

$\phi'(x) = \int_c^d D_1 f(x,t)\,dt$.

Proof. By the mean value theorem, $|f(x,t) - f(x_0,t)| \le |x - x_0|h(t)$, so $\int_c^d f(x,t)dt$ converges by comparison. For $c < d' < d$ let $\phi_{d'}(x) = \int_c^{d'} f(x,t)dt$; then $\phi'_{d'}(x) = \int_c^{d'} D_1 f(x,t)dt$ (Proposition 8.1), and these converge uniformly in x as $d' \to d$, because the tails are bounded by $\int_{d'}^d h$. Also $\phi_{d'}(x_0) \to \phi(x_0)$. Applying Theorem 6.4 along any sequence $d'_n \to d$ gives the result. ∎

Example 8.2

(a) $F(x) = \int_0^\infty e^{-t^2}\cos(xt)\,dt$ satisfies $F'(x) = -\int_0^\infty te^{-t^2}\sin(xt)\,dt = -\frac{x}{2}F(x)$ (integration by parts), so $F(x) = F(0)e^{-x^2/4} = \frac{\sqrt{\pi}}{2}e^{-x^2/4}$.
(b) $\int_0^\infty \frac{\sin x}{x}dx = \frac{\pi}{2}$: for $G(s) = \int_0^\infty e^{-sx}\frac{\sin x}{x}dx$ $(s > 0)$, Theorem 8.3 gives $G'(s) = -\frac{1}{1+s^2}$, and $G(s) \to 0$ as $s \to \infty$, so $G(s) = \frac{\pi}{2} - \arctan s$; continuity at $s = 0$ follows from the second mean value theorem (Corollary 5.25 of Volume I).

8.2 The gamma function

Motivation. The gamma function extends the factorial to real arguments and appears throughout probability and analysis. The theorems of the previous section give its basic properties, and the beta integral and Stirling's formula follow.

For $x > 0$ let $\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}\,dt$, convergent since $t^{x-1}e^{-t} \le t^{x-1}$ near 0 and $\le Ct^{-2}$ near ∞. By Theorem 8.3 on compact subintervals of $(0,\infty)$, Γ is C^∞ with $\Gamma^{(n)}(x) = \int_0^\infty t^{x-1}(\log t)^n e^{-t}dt$. Integration by parts gives $\Gamma(x+1) = x\Gamma(x)$, and $\Gamma(1) = 1$, so $\Gamma(n+1) = n!$.

A function $f : I \to \mathbb{R}$ is **convex** if $f((1-\lambda)x + \lambda y) \le (1-\lambda)f(x) + \lambda f(y)$ for $x, y \in I$, $0 \le \lambda \le 1$.

Proposition 8.4 (8.2.1)

(a) A continuous $f : I \to \mathbb{R}$ with $f(\frac{x+y}{2}) \le \frac{f(x)+f(y)}{2}$ for all $x, y \in I$ is convex.
(b) A convex f satisfies, for $s < t < u$ in I,

$$\frac{f(t) - f(s)}{t - s} \le \frac{f(u) - f(s)}{u - s} \le \frac{f(u) - f(t)}{u - t}.$$

Proof. (a) By induction on n, the convexity inequality holds for dyadic $\lambda = k/2^n$: if $\lambda = \frac{\lambda_1+\lambda_2}{2}$ with dyadic λ_i of level $n-1$, then $(1-\lambda)x + \lambda y$ is the midpoint of the points for λ_1, λ_2. For general λ, approximate by dyadic $\lambda_n \to \lambda$ and use continuity.

(b) Write $t = (1-\lambda)s + \lambda u$ with $\lambda = \frac{t-s}{u-s}$; then $f(t) \le (1-\lambda)f(s) + \lambda f(u)$, which rearranges to each of the two inequalities. ∎

$\log\Gamma$ is convex: by Hölder's inequality with exponents $1/\lambda$, $1/(1-\lambda)$ (or Cauchy–Schwarz for the midpoint and Proposition 8.4(a)),

$$\Gamma\Big(\frac{x+y}{2}\Big) = \int_0^\infty t^{\frac{x-1}{2}}e^{-t/2}\,t^{\frac{y-1}{2}}e^{-t/2}\,dt \le \Gamma(x)^{1/2}\Gamma(y)^{1/2}.$$

Theorem 8.5 (8.2.2, Bohr–Mollerup)

If $f : (0,\infty) \to (0,\infty)$ satisfies (a) $f(x+1) = xf(x)$, (b) $f(1) = 1$, (c) $\log f$ is convex, then $f = \Gamma$.

Proof. Γ has these properties, so it suffices to show f is determined by them; by (a) it suffices to determine f on $(0,1]$. Let $\phi = \log f$, $0 < x \le 1$, $n \in \mathbb{N}$. By (a), (b), $\phi(n+1) = \log n!$. Apply Proposition 8.4(b) to the points $n < n+1 < n+1+x$ and $n+1 < n+1+x < n+2$:

$$\log n \le \frac{\phi(n+1+x) - \phi(n+1)}{x} \le \log(n+1).$$

By (a), $\phi(n+1+x) = \phi(x) + \log[x(x+1)\cdots(x+n)]$. Hence

$$0 \le \phi(x) - \log \frac{n!\, n^x}{x(x+1)\cdots(x+n)} \le x \log\Big(1 + \frac{1}{n}\Big) \to 0,$$

so $f(x) = \lim_n \frac{n!\, n^x}{x(x+1)\cdots(x+n)}$ (Gauss's formula) is determined. ∎

Example 8.3

The **beta function** $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$ satisfies $B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$: for fixed y, $f(x) = \Gamma(x+y)B(x,y)/\Gamma(y)$ satisfies (a), (b), (c) of Theorem 8.5. In particular $\Gamma(\frac12)^2 = B(\frac12,\frac12) = \pi$, so $\int_{-\infty}^{\infty} e^{-x^2}dx = \Gamma(\frac12) = \sqrt{\pi}$. Stirling's formula $\Gamma(x+1) \sim \sqrt{2\pi x}\, x^x e^{-x}$ follows from the substitution $t = x(1+u)$ and Laplace's method.

8.3 Selected exercises

Motivation. These problems from the textbook's exercise set apply differentiation under the integral sign to evaluate classical integrals, and use the functional equation of Γ and termwise integration to compute with the gamma function.

Exercise 8.4.1 (The Gaussian integral)

Let

$$f(x) = \Big(\int_0^x e^{-t^2}dt\Big)^2, \qquad g(x) = \int_0^1 \frac{e^{-x^2(t^2+1)}}{t^2+1}\,dt.$$

Compute $(f+g)'$, show that $f(x) + g(x) = \frac{\pi}{4}$, and deduce $\lim_{A\to\infty} \int_0^A e^{-t^2}dt = \frac12\sqrt{\pi}$.

Solution.

Step 1. By the fundamental theorem of calculus, $f'(x) = 2e^{-x^2}\int_0^x e^{-t^2}dt$.

Step 2. The integrand $h(x,t) = e^{-x^2(t^2+1)}/(t^2+1)$ is continuous on $\mathbb{R}\times[0,1]$ with $D_1h(x,t) = -2xe^{-x^2(t^2+1)}$ continuous. By Proposition 8.1(b) on every bounded interval,

$$g'(x) = -2e^{-x^2}\int_0^1 xe^{-x^2t^2}dt = -2e^{-x^2}\int_0^x e^{-u^2}du,$$

by the substitution $u = xt$ (valid for every real x, including $x \le 0$).

Step 3. So $(f+g)' = 0$, and $f+g$ is constant, equal to $f(0)+g(0) = 0 + \int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4}$.

Step 4. Since $0 \le g(x) \le e^{-x^2}\int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4}e^{-x^2}$, $g(x) \to 0$ and so $f(x) \to \frac{\pi}{4}$ as $x \to \infty$. As $\int_0^x e^{-t^2}dt \ge 0$ for $x \ge 0$, taking square roots gives $\int_0^A e^{-t^2}dt \to \frac{\sqrt{\pi}}{2}$. ∎

Exercise 8.4.4 (A Laplace-type integral)

For $x \in \mathbb{R}$ let

$$f(x) = \int_0^\infty \frac{\cos tx}{1+t^2}\,dt, \qquad g(x) = \int_0^\infty \frac{\sin tx}{t(1+t^2)}\,dt.$$

(a) Show that g is differentiable on $\mathbb{R}$ and $g' = f$.
(b) Show that f is continuous on $\mathbb{R}$.
(c) Show that f is differentiable on $(0,\infty)$ and $g''(x) - g(x) + \frac{\pi}{2} = 0$ for $x > 0$.
(d) Show that $f(x) = \frac{\pi}{2}e^{-|x|}$ for every $x \in \mathbb{R}$.

Solution.

(a) Let $k(x,t) = \frac{\sin tx}{t(1+t^2)}$ for $t > 0$ and $k(x,0) = x$; then k is continuous on $\mathbb{R} \times [0,\infty)$ (as $\frac{\sin tx}{t} \to x$ locally uniformly in x), and $D_1k(x,t) = \frac{\cos tx}{1+t^2}$ is continuous with $|D_1k| \le \frac{1}{1+t^2}$, which is integrable on $[0,\infty)$. Since $|\sin tx| \le |tx|$, $|k(x,t)| \le \frac{|x|}{1+t^2}$, so $g(x)$ converges. Theorem 8.3 on every bounded interval gives $g'(x) = \int_0^\infty \frac{\cos tx}{1+t^2}\,dt = f(x)$.

(b) $\left|\frac{\cos tx}{1+t^2}\right| \le \frac{1}{1+t^2}$, so f is continuous by Theorem 8.2. Also $|f| \le \frac{\pi}{2}$ and $f(0) = \frac{\pi}{2}$.

(c) Fix $x > 0$.

Step 1. Since $\frac{1}{t(1+t^2)} = \frac{1}{t} - \frac{t}{1+t^2}$ and $\int_0^\infty \frac{\sin tx}{t}\,dt = \frac{\pi}{2}$ (substitute $u = tx$ in Example 8.2(b)), the integral $\int_0^\infty \frac{t\sin tx}{1+t^2}dt$ converges as a difference of convergent integrals, and the substitution $u = tx$ gives

$$g(x) = \frac{\pi}{2} - H(x), \qquad H(x) = \int_0^\infty \frac{u\sin u}{x^2+u^2}\,du.$$

Step 2. Integrating by parts, with $\frac{d}{du}\frac{u}{x^2+u^2} = \frac{x^2-u^2}{(x^2+u^2)^2}$ and boundary terms $-\frac{u\cos u}{x^2+u^2}\Big|_0^\infty = 0$,

$$H(x) = \int_0^\infty \frac{(x^2-u^2)\cos u}{(x^2+u^2)^2}\,du.$$

Step 3. The substitution $u = tx$ also gives $f(x) = \int_0^\infty \frac{x\cos u}{x^2+u^2}\,du$. Here $\frac{\partial}{\partial x}\frac{x\cos u}{x^2+u^2} = \frac{(u^2-x^2)\cos u}{(x^2+u^2)^2}$, bounded in absolute value by $\frac{1}{x^2+u^2} \le \frac{1}{a^2+u^2}$ for $x \ge a > 0$, which is integrable on $[0,\infty)$. By Theorem 8.3 on (a,b), for any $0 < a < b$, f is differentiable on $(0,\infty)$ and, by Step 2,

$$f'(x) = \int_0^\infty \frac{(u^2-x^2)\cos u}{(x^2+u^2)^2}\,du = -H(x) = g(x) - \frac{\pi}{2}.$$

Step 4. With $g' = f$ from (a), $g''(x) = f'(x) = g(x) - \frac{\pi}{2}$.

(d)

Step 1. Let $y = g - \frac{\pi}{2}$ on $(0,\infty)$, so $y'' = y$. Put $w = y' + y$; then $w' = y'' + y' = w$, so $(e^{-x}w)' = 0$ and $w = Ce^x$. Then $(e^x y)' = e^x w = Ce^{2x}$, so $y = Ae^x + Be^{-x}$ for constants A, B.

Step 2. $f = y' = Ae^x - Be^{-x}$ is bounded on $(0,\infty)$ by (b), so $A = 0$ and $f(x) = -Be^{-x}$ for $x > 0$. Continuity of f at 0 and $f(0) = \frac{\pi}{2}$ give $-B = \frac{\pi}{2}$.

Step 3. Hence $f(x) = \frac{\pi}{2}e^{-x}$ for $x \ge 0$, and since f is even, $f(x) = \frac{\pi}{2}e^{-|x|}$ for all x. ■

Exercise 8.4.6 (Gamma at half-integers)

Show that $\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)!}{n!\,4^n}\sqrt{\pi}$ for $n = 0, 1, 2, \ldots$.

Solution. For $n = 0$ this is $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (Example 8.3). If it holds for n, then $\Gamma(x+1) = x\Gamma(x)$ gives

$$\Gamma\Big(n + \frac{3}{2}\Big) = \frac{2n+1}{2} \cdot \frac{(2n)!}{n!\,4^n}\sqrt{\pi} = \frac{(2n+2)(2n+1)(2n)!}{4(n+1)\,n!\,4^n}\sqrt{\pi} = \frac{(2n+2)!}{(n+1)!\,4^{n+1}}\sqrt{\pi},$$

which is the claim for $n+1$. ■

Exercise 8.4.7 (A series for Γ)

Show that for every $x > 0$,

$$\Gamma(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \frac{1}{n+x} + \sum_{n=0}^{\infty} \Big(\frac{1}{n!}\int_1^{\infty} t^{-1}e^{-t}(\log t)^n dt\Big)x^n.$$

Solution. Split $\Gamma(x) = \int_0^1 t^{x-1}e^{-t}dt + \int_1^{\infty} t^{x-1}e^{-t}dt$.

Step 1. (The integral over $(0,1]$.) Let $R_N(t) = e^{-t} - \sum_{n\le N} \frac{(-t)^n}{n!}$. On $[0,1]$, $|R_N(t)| \le \sum_{n>N} \frac{1}{n!} =: \rho_N \to 0$. Since $\int_0^1 t^{x-1}dt = \frac{1}{x}$ and $\int_0^1 t^{n+x-1}dt = \frac{1}{n+x}$,

$$\Big|\int_0^1 t^{x-1}e^{-t}dt - \sum_{n\le N} \frac{(-1)^n}{n!(n+x)}\Big| = \Big|\int_0^1 t^{x-1}R_N(t)\,dt\Big| \le \frac{\rho_N}{x} \to 0.$$

Step 2. (The integral over $[1,\infty)$: setup.) For $t \ge 1$, $t^{x-1}e^{-t} = t^{-1}e^{-t}e^{x\log t} = \sum_n \phi_n(t)$ with $\phi_n(t) = \frac{x^n}{n!}t^{-1}e^{-t}(\log t)^n \ge 0$. Each $\int_1^{\infty}\phi_n$ converges, as $\phi_n \le t^{x-1}e^{-t}$.

Step 3. (Upper bound.) For every N, $\sum_{n\le N}\phi_n \le t^{x-1}e^{-t}$, so $\sum_{n\le N}\int_1^{\infty}\phi_n \le \int_1^{\infty} t^{x-1}e^{-t}dt$. The series $\sum_n \int_1^{\infty}\phi_n$ of nonnegative terms therefore converges, with sum at most $\int_1^{\infty} t^{x-1}e^{-t}dt$.

Step 4. (Lower bound.) For $c > 1$, $\log t \le \log c$ on $[1,c]$, so $\sum \phi_n$ converges uniformly on $[1,c]$ (M-test, Corollary 6.3), and by Proposition 6.5

$$\int_1^c t^{x-1}e^{-t}dt = \sum_n \int_1^c \phi_n \le \sum_n \int_1^{\infty}\phi_n.$$

Letting $c \to \infty$ and combining with Step 3, $\int_1^{\infty} t^{x-1}e^{-t}dt = \sum_n \int_1^{\infty}\phi_n$, which is the second series. Adding Step 1 gives the formula. ■

Chapter 9. Fourier Series

9.1 Fourier coefficients

Motivation. If a periodic function is a trigonometric series, its coefficients must be given by integrals against $\cos nx$ and $\sin nx$. We study these coefficients for arbitrary integrable functions: they are the best L^2 approximation and decay to zero.

In an inner product space X, a family $\{\phi_n : n \in \mathbb{Z}\}$ is **orthonormal** if $\langle \phi_m, \phi_n \rangle = \delta_{mn}$. In $\mathcal{R}^2[-\pi,\pi]$ with $\langle f, g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f\bar{g}$, the functions $e_n(x) = e^{inx}$ are orthonormal. For $f \in \mathcal{R}^1[-\pi,\pi]$ (extended 2π-periodically) the **Fourier coefficients** are $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt$, the **Fourier series** is $\sum \hat{f}(n)e^{inx}$, and $s_n(f;x) = \sum_{|k|\le n} \hat{f}(k)e^{ikx}$.

Theorem 9.1 (9.1.1, best approximation and Bessel's inequality)

Let $\{\phi_n\}$ be orthonormal in X, $x \in X$, $c_n = \langle x, \phi_n\rangle$. For all complex z_n,

$$\Big\| x - \sum_{n=-N}^{N} c_n\phi_n \Big\| \le \Big\| x - \sum_{n=-N}^{N} z_n\phi_n \Big\|,$$

with equality iff $z_n = c_n$ for all $|n| \le N$. Moreover $\sum_{n\in\mathbb{Z}} |c_n|^2 \le \|x\|^2$.

Proof. Let $s = \sum_{|n|\le N} c_n\phi_n$ and $t = \sum z_n\phi_n$. Then $\langle x-s, \phi_n\rangle = 0$ for $|n| \le N$, so $x - s \perp s - t$, and $\|x-t\|^2 = \|x-s\|^2 + \|s-t\|^2 = \|x-s\|^2 + \sum|c_n - z_n|^2$. With $t = 0$: $\|x\|^2 = \|x-s\|^2 + \sum_{|n|\le N}|c_n|^2$, which gives Bessel's inequality. ∎

Proposition 9.2 (9.1.2)

For $f \in \mathcal{R}^1[-\pi,\pi]$:
(a) if f is real, $\hat{f}(-n) = \overline{\hat{f}(n)}$;
(b) if f is even, $\hat{f}(-n) = \hat{f}(n)$;
(c) if f is odd, $\hat{f}(-n) = -\hat{f}(n)$.

Proof. (a) Conjugate the integral. (b), (c) Substitute $t \mapsto -t$. ∎

In real form,

$$s_n(f;x) = \frac{a_0}{2} + \sum_{k\le n}(a_k \cos kx + b_k \sin kx)$$

with $a_k = \frac{1}{\pi}\int f(t)\cos kt\,dt$, $b_k = \frac{1}{\pi}\int f(t)\sin kt\,dt$; even functions have cosine series and odd functions sine series.

Example 9.1

$f(x) = x$ on $(-\pi,\pi)$: $\hat{f}(n) = \frac{(-1)^n i}{n}$ $(n \ne 0)$, so $x \sim 2\sum(-1)^{n+1}\frac{\sin nx}{n}$. $f(x) = |x|$: $|x| \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k \text{ odd}} \frac{\cos kx}{k^2}$, and at $x = 0$ this gives $\sum_{k \text{ odd}} k^{-2} = \pi^2/8$, hence $\sum n^{-2} = \pi^2/6$ (once convergence to f is proved, Corollary 9.7).

9.2 Pointwise convergence

Motivation. Does the Fourier series of f converge back to f? Not always, but it does at points where f is reasonably smooth. The Dirichlet kernel reduces the question to an integral, and the Riemann–Lebesgue lemma handles it.

Theorem 9.3 (9.2.1, Riemann–Lebesgue lemma)

If $\int_I |f| < \infty$ on an interval I, then for every $\beta \in \mathbb{R}$, $\lim_{\alpha\to\infty} \int_I f(t)\sin(\alpha t+\beta)\,dt = 0$.

Proof. For $f = \chi_{[a,b]}$, $|\int_a^b \sin(\alpha t+\beta)dt| \le 2/\alpha$; by linearity the claim holds for step functions. For general f and ϵ, choose a step function s with $\int_I |f-s| < \epsilon$ (Theorem 7.16); then $|\int f\sin(\alpha t+\beta)| \le \epsilon + |\int s\sin(\alpha t+\beta)| < 2\epsilon$ for large α. ∎

In particular $\hat{f}(n) \to 0$ as $|n| \to \infty$ (Theorem 9.21). The partial sums are integrals against the **Dirichlet kernel**:

$$s_n(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_n(t)\,dt, \qquad D_n(t) = \sum_{|k|\le n} e^{ikt} = \frac{\sin(n+\frac12)t}{\sin\frac{t}{2}}, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} D_n = 1.$$

Since D_n is even, $s_n(f;x) = \frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)]D_n(t)\,dt$.

Theorem 9.4 (9.2.2, Riemann localization)

Let $0 < \delta < \pi$ and $\int_{-\pi}^{\pi}|f| < \infty$. Then $\lim_n s_n(f;x)$ exists iff $\lim_n \frac{1}{\pi}\int_0^{\delta}[f(x+t)+f(x-t)]\frac{\sin(n+\frac12)t}{t}\,dt$ exists, and then they are equal. So convergence of the Fourier series at x depends only on f near x.

Proof. $\frac{1}{\sin(t/2)} - \frac{2}{t}$ is bounded on $(0,\pi]$, so by Theorem 9.3, $\frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)](\frac{1}{\sin(t/2)} - \frac{2}{t})\sin(n+\frac12)t\,dt \to 0$; and $\frac{2}{t}$ is bounded on $[\delta,\pi]$, so the integral over $[\delta,\pi]$ with kernel $\frac{2\sin(n+\frac12)t}{t}$ also tends to 0. ∎

We use $\lim_{\lambda\to\infty}\int_0^{\delta}\frac{\sin\lambda t}{t}\,dt = \lim\int_0^{\lambda\delta}\frac{\sin u}{u}du = \frac{\pi}{2}$ (Example 8.2).

Lemma 9.5 (9.2.3)

If $g(0+)$ exists and $\int_0^{\delta}\frac{|g(t)-g(0+)|}{t}\,dt < \infty$, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^{\delta} g(t)\frac{\sin\lambda t}{t}\,dt = g(0+)$.

Proof. Write $g(t) = g(0+) + (g(t)-g(0+))$. The first part gives $g(0+)\cdot\frac{2}{\pi}\cdot\frac{\pi}{2}$ in the limit; the second tends to 0 by Theorem 9.3 applied to the integrable function $(g(t)-g(0+))/t$. ∎

Theorem 9.6 (9.2.4, Dini's test)

Let $f \in \mathcal{R}^1[-\pi,\pi]$ and $x \in [-\pi,\pi]$. Suppose (a) $f(x+)$ and $f(x-)$ exist, and (b) $\int_0^{\delta}\frac{|f(x+t)+f(x-t)-f(x+)-f(x-)|}{t}\,dt < \infty$. Then $s_n(f;x) \to \frac12[f(x+)+f(x-)]$.

Proof. Apply Theorem 9.4 and Lemma 9.5 to $g(t) = \frac12[f(x+t)+f(x-t)]$, with $\lambda = n+\frac12$. ∎

Corollary 9.7 (9.2.5)

If f satisfies a Lipschitz condition at x, $|f(x+t)-f(x)| \le Mt$ for small $|t|$, then $s_n(f;x) \to f(x)$.

Proof. The integrand in (b) is at most $2M$. ∎

Corollary 9.8 (9.2.6)

If f is differentiable at x, then $s_n(f;x) \to f(x)$.

Proof. Differentiability at x implies $|f(x+t)-f(x)| \le (|f'(x)|+1)|t|$ for small t. ∎

Lemma 9.9 (9.2.7)

If $g:[0,\delta]\to\mathbb{R}$ is of bounded variation, then $\lim_{\lambda\to\infty} \frac{2}{\pi}\int_0^\delta g(t)\frac{\sin \lambda t}{t}\,dt = g(0+)$.

Proof. By Jordan decomposition we may assume g increasing, and subtracting a constant, $g(0+)=0$. Given ϵ, choose $h\in(0,\delta)$ with $g(h)<\epsilon$. On $[h,\delta]$, $g(t)/t$ is integrable, so that part tends to 0 by Theorem 9.3. On $(0,h]$, by the second mean value theorem (Corollary 5.25 of Volume I, applied on $[\eta,h]$ and letting $\eta\to 0$), $\int_0^h g(t)\frac{\sin\lambda t}{t}dt = g(h-)\int_c^h \frac{\sin\lambda t}{t}dt$ for some $c\in[0,h]$. Since $|\int_a^b \frac{\sin u}{u}du| \le K$ for all $0\le a<b$ (a bounded quantity, as $\int_0^\infty \frac{\sin u}{u}$ converges), this part is at most ϵK. ∎

Theorem 9.10 (9.2.8, Dirichlet–Jordan)

If $f\in\mathcal{R}^1[-\pi,\pi]$ is of bounded variation on some interval $[x-\delta,x+\delta]$, then $s_n(f;x)\to \frac{1}{2}[f(x+)+f(x-)]$.

Proof. $g(t)=\frac{1}{2}[f(x+t)+f(x-t)]$ is of bounded variation on $[0,\delta]$ with $g(0+)=\frac{1}{2}[f(x+)+f(x-)]$; apply Theorem 9.4 and Lemma 9.9. ∎

Example 9.2

For $f(x)=x$ on $(-\pi,\pi)$ the series converges to x inside and to 0 at $\pm\pi$; at $x=\pi/2$ it gives $\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\cdots$. Near the jump the partial sums overshoot by about 9% of the jump (Gibbs phenomenon). There exist continuous functions whose Fourier series diverge at a point (du Bois-Reymond).

9.3 Fejér's theorem

Motivation. Even when a Fourier series diverges, averaging its partial sums converges uniformly to any continuous periodic function. This shows that a continuous function is determined by its Fourier coefficients and gives another proof of Weierstrass approximation.

The **Cesàro means** are $\sigma_n(f;x)=\frac{1}{n+1}\sum_{k=0}^n s_k(f;x)=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)K_n(t)\,dt$ with the **Fejér kernel**

$$K_n(t)=\frac{1}{n+1}\sum_{k=0}^{n} D_k(t)=\frac{1}{n+1}\Big(\frac{\sin\frac{(n+1)t}{2}}{\sin\frac{t}{2}}\Big)^2.$$

It satisfies $K_n\ge 0$, $\frac{1}{2\pi}\int K_n=1$, and $K_n(t)\le \frac{1}{(n+1)\sin^2(\delta/2)}\to 0$ uniformly on $\delta\le|t|\le\pi$. (The formula follows from $\sum_{k\le n}\sin(k+\frac{1}{2})t\cdot\sin\frac{t}{2}=\frac{1}{2}\sum(\cos kt-\cos(k+1)t)=\sin^2\frac{(n+1)t}{2}$.)

Theorem 9.11 (9.3.1, Fejér)

If $f \in \mathcal{R}^1[-\pi,\pi]$ and $f(x\pm)$ exist, then $\sigma_n(f;x) \to \frac{1}{2}[f(x+)+f(x-)]$. If f is continuous and 2π-periodic, $\sigma_n(f) \to f$ uniformly.

Proof. Since K_n is even, $\sigma_n(f;x) - s = \frac{1}{2\pi}\int_0^\pi [f(x+t)+f(x-t)-2s]K_n(t)\,dt$ with $s = \frac{1}{2}[f(x+)+f(x-)]$. Given ϵ, choose δ with $|f(x+t)+f(x-t)-2s| < \epsilon$ for $0 < t < \delta$. The integral over $(0,\delta)$ is at most $\epsilon \cdot \frac{1}{2\pi}\int_0^\pi K_n \le \epsilon/2$, and that over $[\delta,\pi]$ is at most $\frac{1}{(n+1)\sin^2(\delta/2)} \cdot \frac{1}{2\pi}\int |f(x+t)+f(x-t)-2s| \to 0$. For continuous periodic f, δ can be chosen uniformly in x and f is bounded, giving uniform convergence. ■

So every continuous periodic function is a uniform limit of trigonometric polynomials (the trigonometric form of Weierstrass's theorem), and a continuous periodic function is determined by its Fourier coefficients.

Lemma 9.12 (9.3.2)

If $s_n \to s$, then $\sigma_n = \frac{1}{n+1}\sum_{k\le n} s_k \to s$.

Proof. We may assume $s = 0$. Given ϵ, choose N with $|s_n| < \epsilon$ for $n \ge N$. For $n > N$, $|\sigma_n| \le \frac{1}{n+1}\sum_{k<N}|s_k| + \epsilon < 2\epsilon$ for large n. ■

Corollary 9.13 (9.3.3)

If the Fourier series of a continuous periodic f converges at x, its sum is $f(x)$.

Proof. By Lemma 9.12 the limit equals $\lim \sigma_n(f;x) = f(x)$ (Theorem 9.11). ■

Corollary 9.14 (9.3.4)

If $\int_{-\pi}^{\pi}|f|^2 < \infty$, then $\lim_n \|f - s_n(f)\|_2 = 0$.

Proof. Given ϵ, choose a continuous g vanishing near $\pm\pi$ (so continuous as a periodic function) with $\|f-g\|_2 < \epsilon$ (Theorem 7.14 and Lemma 7.13), and by Theorem 9.11 a trigonometric polynomial $T = \sigma_m(g)$ of degree m with $\|g - T\|_{\sup} < \epsilon$, so $\|g-T\|_2 < \epsilon$. For $n \ge m$, Theorem 9.1 gives $\|f - s_n(f)\|_2 \le \|f - T\|_2 < 2\epsilon$. ■

Corollary 9.15 (9.3.5, Parseval's identity)

If $I = [a, a+2\pi]$, $f \in \mathcal{R}^2(I)$ and $c_n = \frac{1}{2\pi}\int_I f(t)e^{-int}dt$, then $\frac{1}{2\pi}\int_I |f|^2 = \sum_{n\in\mathbb{Z}}|c_n|^2$; more generally $\frac{1}{2\pi}\int_I f\bar{g} = \sum c_n\overline{d_n}$ for $g \in \mathcal{R}^2(I)$ with coefficients d_n.

Proof. By periodicity the coefficients do not depend on a. From the proof of Theorem 9.1, $\|f\|_2^2 = \|f - s_n(f)\|_2^2 + \sum_{|k|\le n}|c_k|^2$; let $n \to \infty$ (Corollary 9.14). The polarized form follows from $4\langle f, g\rangle = \sum_{k=0}^{3} i^k \|f + i^k g\|^2$. ■

Example 9.3

Parseval for $f(x) = x$ gives $\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{3} = 2\sum_{n\ge1}\frac{1}{n^2}$, again $\sum n^{-2} = \pi^2/6$.

9.4 Differentiation and integration of Fourier series

Motivation. Fourier series can always be integrated term by term, and differentiated term by term under smoothness assumptions. These rules make Fourier series a practical tool for differential equations.

Theorem 9.16 (9.4.1, termwise integration)

If $f \in \mathcal{R}^2(I)$ has Fourier series $\sum \hat f(n)e^{int}$, then for $a, x \in I$, $\int_a^x f(t)\,dt = \sum_n \hat f(n) \int_a^x e^{int}dt$, the series converging (uniformly in x).

Proof. By Cauchy–Schwarz, $|\int_a^x (f - s_n(f))| \le \sqrt{2\pi}\Big(\int_I |f - s_n(f)|^2\Big)^{1/2} \to 0$ by Corollary 9.14. ∎

Note that the integrated series converges whether or not the Fourier series of f converges. If f is periodic and C^1, integration by parts gives $\widehat{f'}(n) = in\hat f(n)$; since $\sum |n\hat f(n)|^2 < \infty$, Cauchy–Schwarz gives $\sum |\hat f(n)| < \infty$, so the Fourier series converges uniformly (to f, by Corollary 9.13).

A sequence $a : \mathbb{Z} \to \mathbb{C}$ is **rapidly decreasing** if $|n|^k a(n)$ is bounded for every k.

Theorem 9.17 (9.4.2)

The Fourier coefficients of a C^∞ periodic function are rapidly decreasing. Conversely, if a is rapidly decreasing, $f(x) = \lim_n \sum_{|k|\le n} a(k)e^{ikx}$ defines a C^∞ periodic function with $\hat f(n) = a(n)$.

Proof. Repeated integration by parts gives $\widehat{f^{(k)}}(n) = (in)^k \hat f(n)$, so $|n^k \hat f(n)| \le \frac{1}{2\pi}\int |f^{(k)}|$. Conversely the series and all its term-by-term derivatives $\sum (ik)^j a(k) e^{ikx}$ converge uniformly by the M-test (as $\sum |k|^j |a(k)| < \infty$), so f is C^∞ (Theorem 6.4); integrating the uniformly convergent series against e^{-inx} term by term gives $\hat f(n) = a(n)$. ∎

Lemma 9.18 (9.4.3)

If $f : (a, b) \to \mathbb{R}$ is analytic and $K \subset (a, b)$ is compact, there are $C, R > 0$ with $|f^{(n)}(x)| \le Cn!R^n$ for $x \in K$, $n = 1, 2, \dots$.

Proof. Fix $c \in K$; near c, $f(x) = \sum_k a_k (x - c)^k$ with positive radius, so there are $\rho > 0$, M with $|a_k|\rho^k \le M$. For $|x - c| \le \rho/2$, termwise differentiation (Theorem 6.12) gives

$$|f^{(n)}(x)| \le \sum_{k \ge n} \frac{k!}{(k-n)!} |a_k| \Big(\frac{\rho}{2}\Big)^{k-n} \le M\rho^{-n} n! \sum_{j \ge 0} \binom{n+j}{n} 2^{-j} = 2M\, n! \Big(\frac{2}{\rho}\Big)^n.$$

Finitely many such neighbourhoods cover K; take the largest constants. ∎

Lemma 9.19 (9.4.4)

If $c \ge 0$, $R > 0$, $n \ge 1$ and $c \le k!\,(R/n)^k$ for all $k = 0, 1, 2, \dots$, then $c \le (R + 1)e^{-n/(R+1)}$.

Proof. If $c > 0$, then $\frac{1}{k!}\big(\frac{n}{R+1}\big)^k \le \frac{1}{c}\big(\frac{R}{R+1}\big)^k$ for every k; summing, $e^{n/(R+1)} \le \frac{R+1}{c}$. ∎

Theorem 9.20 (9.4.5)

If $f:\mathbb{R}\to\mathbb{C}$ is periodic and analytic, then $\limsup_{|n|\to\infty}|\hat f(n)|^{1/|n|}<1$.

Proof. By Lemma 9.18 on $K=[-\pi,\pi]$, $\|f^{(k)}\|_{\sup}\le Ck!R^k$ for $k\ge 1$ (and enlarging C, also for $k=0$). Then $|\hat f(n)|=|\widehat{f^{(k)}}(n)|/|n|^k\le Ck!(R/|n|)^k$ for all k, so by Lemma 9.19, $|\hat f(n)|\le C(R+1)e^{-|n|/(R+1)}$, and $\limsup|\hat f(n)|^{1/|n|}\le e^{-1/(R+1)}<1$. ∎

Conversely, such geometric decay of the coefficients makes f analytic (the series extends to a holomorphic function near the unit circle).

Theorem 9.21 (9.4.6)

If $\int_{-\pi}^{\pi}|f|<\infty$, then $\lim_{|n|\to\infty}\hat f(n)=0$.

Proof. Theorem 9.3 applied to the real and imaginary parts of $\int f(t)(\cos nt - i\sin nt)\,dt$. ∎

The decay can be arbitrarily slow, but not every sequence tending to 0 is a sequence of Fourier coefficients:

Proposition 9.22 (9.4.7)

If $f\in\mathcal{R}^1[-\pi,\pi]$ is odd, then the partial sums of $\sum_{n\ge1}\hat f(n)/n$ are bounded.

Proof. For odd f, $\hat f(n)=-\frac{i}{\pi}\int_0^\pi f(t)\sin nt\,dt$, so $\sum_{n\le N}\frac{\hat f(n)}{n}=-\frac{i}{\pi}\int_0^\pi f(t)\sum_{n\le N}\frac{\sin nt}{n}\,dt$. The sums $\sum_{n\le N}\frac{\sin nt}{n}$ are bounded by $1+2\pi$ on $(0,\pi]$: the terms with $n\le 1/t$ are at most t each, and for $n>1/t$, summation by parts with (‡) of Chapter 6 gives at most $2\cdot t\cdot\pi/t$. Hence the partial sums are bounded by $\frac{1+2\pi}{\pi}\int_0^\pi|f|$. ∎

So $\sum_{n\ge2}\frac{\sin nx}{\log n}$, which converges everywhere (Theorem 6.20), is not the Fourier series of any $f\in\mathcal{R}^1$, since $\sum\frac{1}{n\log n}=\infty$.

9.5 Selected exercises

Motivation. These problems from the textbook's exercise set explain why partial sums can misbehave while Cesàro means cannot, characterise the averaging methods that preserve limits, and show that bounded variation forces the Fourier coefficients to decay like $1/|n|$.

Exercise 9.5.3 (Lebesgue constants)

Show that $\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n(t)|\,dt\to\infty$ as $n\to\infty$; in fact it is at least $\frac{4}{\pi^2}(1+\frac12+\cdots+\frac1n)$.

Solution.

Step 1. D_n is even and $0<\sin\frac{t}{2}\le\frac{t}{2}$ on $(0,\pi]$, so

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n|=\frac{1}{\pi}\int_0^\pi\frac{|\sin(n+\frac12)t|}{\sin\frac{t}{2}}\,dt\ge\frac{2}{\pi}\int_0^\pi\frac{|\sin(n+\frac12)t|}{t}\,dt=\frac{2}{\pi}\int_0^{(n+\frac12)\pi}\frac{|\sin u|}{u}\,du,$$

by the substitution $u=(n+\frac12)t$.

Step 2. On $[k\pi,(k+1)\pi]$, $\frac{1}{u}\geq\frac{1}{(k+1)\pi}$ and $\int_{k\pi}^{(k+1)\pi}|\sin u|\,du=2$. Summing over $k=0,\dots,n-1$,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n|\geq\frac{2}{\pi}\sum_{k=0}^{n-1}\frac{2}{(k+1)\pi}=\frac{4}{\pi^2}\sum_{k=1}^{n}\frac{1}{k}\to\infty.$$

By contrast the Fejér kernel satisfies $\frac{1}{2\pi}\int|K_n|=1$ because $K_n\geq 0$; this is why Fejér's theorem holds for all continuous f while the partial sums may diverge (Example 9.2). ■

Exercise 9.5.8 (Regular averaging methods)

Let $a_{mn}\geq 0$ $(m,n\geq 1)$ with $a_{mn}=0$ for $n>m$ and $\sum_{n=1}^{m}a_{mn}=1$ for every m. For a sequence (s_n) put $t_m=\sum_{n=1}^{m}a_{mn}s_n$. Show that the following are equivalent:
(a) for every convergent sequence, $\lim_n s_n=s$ implies $\lim_m t_m=s$;
(b) $\lim_{m\to\infty}a_{mn}=0$ for each n.

Solution.

(a)⇒(b): Fix n_0 and let $s_{n_0}=1$, $s_n=0$ for $n\neq n_0$. Then $s_n\to 0$ and $t_m=a_{mn_0}$, so (a) gives $a_{mn_0}\to 0$.

(b)⇒(a):

Step 1. Since $\sum_n a_{mn}=1$, $t_m-s=\sum_{n\leq m}a_{mn}(s_n-s)$.

Step 2. Given $\epsilon>0$, choose N with $|s_n-s|<\epsilon$ for $n>N$. For $m>N$, using $a_{mn}\geq 0$,

$$|t_m-s|\leq\sum_{n\leq N}a_{mn}|s_n-s|+\epsilon\sum_{N<n\leq m}a_{mn}\leq\sum_{n\leq N}a_{mn}|s_n-s|+\epsilon.$$

Step 3. The first sum has N fixed terms, each tending to 0 as $m\to\infty$ by (b). Hence $\limsup_m|t_m-s|\leq\epsilon$ for every ϵ, i.e. $t_m\to s$.

Taking $a_{mn}=\frac{1}{m}$ for $n\leq m$ recovers Lemma 9.12. ■

Exercise 9.5.13 (Fourier coefficients of functions of bounded variation)

A function $f:[-\pi,\pi]\to\mathbb{C}$ is of bounded variation if $\sup_P\sum_i|f(x_i)-f(x_{i-1})|<\infty$ over all partitions P.
(a) Show that a function of bounded variation is a linear combination of increasing real functions.
(b) Show that for every increasing $f:[-\pi,\pi]\to\mathbb{R}$, $|\hat f(n)|\leq\frac{V_{-\pi}^{\pi}(f)}{\pi|n|}$ for $n=\pm1,\pm2,\dots$.
(c) Show that the Fourier coefficients of a function f of bounded variation satisfy $|\hat f(n)|\leq\frac{C}{|n|}$ $(n\neq 0)$ for some constant $C>0$.

Solution.

(a) Since $|\operatorname{Re}z|\leq|z|$ and $|\operatorname{Im}z|\leq|z|$, the real functions $\operatorname{Re}f$ and $\operatorname{Im}f$ are of bounded variation. By the Jordan decomposition (Theorem 5.16 of Volume I), $\operatorname{Re}f=g_1-h_1$ and $\operatorname{Im}f=g_2-h_2$ with g_j,h_j increasing, so $f=g_1-h_1+ig_2-ih_2$.

(b) Let $V=V_{-\pi}^{\pi}(f)=f(\pi)-f(-\pi)$ and $n\neq 0$.

Step 1. Choose $\theta\in\mathbb{R}$ with $e^{i\theta}\int_{-\pi}^{\pi}f(t)e^{-int}dt\geq 0$. Then

$$2\pi|\hat f(n)|=\operatorname{Re}\Big(e^{i\theta}\int_{-\pi}^{\pi}f(t)e^{-int}dt\Big)=\int_{-\pi}^{\pi}f(t)\cos(nt-\theta)\,dt.$$

Step 2. By the second mean value theorem (Corollary 5.25 of Volume I) there is $c \in [-\pi, \pi]$ with

$$\int_{-\pi}^{\pi} f(t)\cos(nt-\theta)\,dt = f(-\pi)\int_{-\pi}^{c}\cos(nt-\theta)\,dt + f(\pi)\int_{c}^{\pi}\cos(nt-\theta)\,dt.$$

Step 3. Since $\int_{-\pi}^{\pi}\cos(nt-\theta)\,dt = 0$, the right side equals $-V\int_{-\pi}^{c}\cos(nt-\theta)\,dt = -\frac{V}{n}\big(\sin(nc-\theta) - \sin(-n\pi-\theta)\big)$, whose absolute value is at most $\frac{2V}{|n|}$.

Step 4. Hence $2\pi|\hat f(n)| \le \frac{2V}{|n|}$, i.e. $|\hat f(n)| \le \frac{V}{\pi|n|}$.

(c) Write $f = g_1 - h_1 + ig_2 - ih_2$ as in (a). By linearity of $f \mapsto \hat f(n)$ and (b),

$$|\hat f(n)| \le |\hat g_1(n)| + |\hat h_1(n)| + |\hat g_2(n)| + |\hat h_2(n)| \le \frac{C}{|n|},$$

with $C = \frac{1}{\pi}\big(V(g_1)+V(h_1)+V(g_2)+V(h_2)\big)$, where $V(\cdot) = V_{-\pi}^{\pi}(\cdot)$. For instance $f(x) = x$ has $|\hat f(n)| = \frac{1}{|n|}$ (Example 9.1), so the order $\frac{1}{|n|}$ cannot be improved. ∎

Chapter 10. The Lebesgue Integral

10.1 Measurable sets and Lebesgue measure

Motivation. The Riemann integral handles poorly functions with many discontinuities and limits of sequences of functions. Lebesgue's idea is to partition the range instead of the domain, which requires measuring the size of arbitrary sets of the form $\{x : f(x) > c\}$. Lebesgue measure extends length to such sets.

Proposition 10.1 (10.1.1)

Every open $U \subset \mathbb{R}$ is a union of countably many disjoint open intervals, and this representation is unique.

Proof. For $x \in U$ let $I_x = (a_x, b_x)$ with $a_x = \inf\{a : (a, x] \subset U\}$, $b_x = \sup\{b : [x, b) \subset U\}$; $I_x \subset U$ is the largest open interval in U containing x. If $I_x \cap I_y \neq \emptyset$, then $I_x \cup I_y$ is an open interval in U, so $I_x = I_y$. Each I_x contains a rational, and distinct ones are disjoint, so there are countably many. The intervals are the components of U, which gives uniqueness. ■

For $U = \bigsqcup(a_n, b_n)$ put $\lambda(U) = \sum(b_n - a_n)$ (independent of the order, Proposition 2.16 of Volume I), and for any $S \subset \mathbb{R}$ define the **outer measure**

$$\mu(S) = \inf\{\lambda(U) : U \supset S \text{ open}\} \in [0, \infty].$$

Proposition 10.2 (10.1.2)

(a) $S \subset T \Rightarrow \mu(S) \leq \mu(T)$.
(b) If I is an interval with end points $a \leq b$, then $\mu(I) = b - a$.
(c) $\mu(x + S) = \mu(S)$.
(d) $\mu(\bigcup S_n) \leq \sum \mu(S_n)$ for countably many S_n.

Proof. First, if an open interval (a, b) is covered by open intervals J_k, then $b - a \leq \sum |J_k|$: for $a < c < d < b$ finitely many J_k cover the compact $[c, d]$, and by choosing a chain of overlapping intervals from c to d, $d - c \leq \sum |J_k|$; let $c \to a$, $d \to b$. Summing over components (Proposition 6.9), $\lambda(U) \leq \sum \lambda(U_n)$ whenever $U \subset \bigcup U_n$ are open.

(a) is immediate.

(b) For $I = [a, b]$, $\mu(I) \leq \lambda((a - \epsilon, b + \epsilon)) = b - a + 2\epsilon$; and an open $U \supset I$ has a component containing $[a, b]$, so $\lambda(U) \geq b - a$. Other intervals are similar.

(c) Translation preserves λ of open sets.

(d) We may assume all $\mu(S_n) < \infty$. Choose open $U_n \supset S_n$ with $\lambda(U_n) < \mu(S_n) + \epsilon 2^{-n}$; then $\mu(\bigcup S_n) \leq \lambda(\bigcup U_n) \leq \sum \lambda(U_n) < \sum \mu(S_n) + \epsilon$. ■

In particular countable sets have outer measure 0, so $[0, 1]$ is uncountable. A set $E \subset \mathbb{R}$ is **measurable** if

$$\mu(X) = \mu(X \cap E) + \mu(X \cap E^c) \qquad \text{for every } X \subset \mathbb{R};$$

let $\mathfrak{M}$ be the family of measurable sets. By (d), only $\geq$ needs checking, and only for $\mu(X) < \infty$.

Theorem 10.3 (10.1.3)

(a) $\emptyset \in \mathfrak{M}$.
(b) $E \in \mathfrak{M} \Rightarrow E^c \in \mathfrak{M}$.
(c) $E, F \in \mathfrak{M} \Rightarrow E \cup F \in \mathfrak{M}$.
(d) $E_n \in \mathfrak{M} \Rightarrow \bigcup E_n \in \mathfrak{M}$.
(e) If $\mu(E) = 0$, then $E \in \mathfrak{M}$.
(f) Every interval is measurable.
(g) Every open set is measurable.

Proof. (a), (b) are clear from the symmetric definition.

(c) Since $X \cap (E \cup F) = (X \cap E) \cup (X \cap F \cap E^c)$,

$$\mu(X\cap(E\cup F))+\mu(X\cap(E\cup F)^c) \le \mu(X\cap E)+\mu(X\cap E^c\cap F)+\mu(X\cap E^c\cap F^c) = \mu(X\cap E)+\mu(X\cap E^c) = \mu(X),$$

using $F \in \mathfrak{M}$ (for the set $X \cap E^c$) and then $E \in \mathfrak{M}$.

(d) Let $F_n = E_n - (E_1 \cup \cdots \cup E_{n-1})$ (measurable by (b), (c), as $A \cap B = (A^c \cup B^c)^c$), disjoint, and $G_n = F_1 \cup \cdots \cup F_n$, $E = \bigcup E_n = \bigsqcup F_n$. Using $F_n \in \mathfrak{M}$ for the set $X \cap G_n$, $\mu(X \cap G_n) = \mu(X \cap G_{n-1}) + \mu(X \cap F_n)$, so $\mu(X \cap G_n) = \sum_{i \le n} \mu(X \cap F_i)$. As $G_n \in \mathfrak{M}$ and $G_n^c \supset E^c$,

$$\mu(X) = \mu(X \cap G_n) + \mu(X \cap G_n^c) \ge \sum_{i \le n} \mu(X \cap F_i) + \mu(X \cap E^c).$$

Let $n \to \infty$ and use (d) of Proposition 10.2: $\mu(X) \ge \mu(X \cap E) + \mu(X \cap E^c)$.

(e) $\mu(X \cap E) = 0$ and $\mu(X \cap E^c) \le \mu(X)$.

(f) By (b), (d) and (e) it suffices to treat $I = (a, \infty)$. Given X and ϵ, take open $U \supset X$ with $\lambda(U) < \mu(X)+\epsilon$. The open sets $U\cap(a,\infty), U\cap(-\infty,a)$ satisfy $\lambda(U\cap(a,\infty))+\lambda(U\cap(-\infty,a)) \le \lambda(U)$ (each component of U is split into two intervals), so $\mu(X \cap I) + \mu(X \cap I^c) \le \lambda(U \cap (a, \infty)) + \lambda(U \cap (-\infty, a)) + \mu(\{a\}) < \mu(X) + \epsilon$.

(g) Proposition 10.1, (f) and (d). ■

So $\mathfrak{M}$ is a σ-algebra containing all Borel sets. The **Lebesgue measure** is $m = \mu|_{\mathfrak{M}}$.

Theorem 10.4 (10.1.4, countable additivity)

If $E_n \in \mathfrak{M}$ are disjoint, then $m(\bigsqcup E_n) = \sum m(E_n)$.

Proof. In the proof of Theorem 10.3(d) take $X = E = \bigsqcup E_n$ (so $F_n = E_n$): $m(E) \ge \sum m(E_n)$; the reverse is Proposition 10.2(d). ■

Consequences: if $E_1 \subset E_2 \subset \cdots$, then $m(\bigcup E_n) = \lim m(E_n)$ (write the union as $\bigsqcup(E_n - E_{n-1})$); if $E_1 \supset E_2 \supset \cdots$ and $m(E_1) < \infty$, then $m(\bigcap E_n) = \lim m(E_n)$. By definition of μ, every measurable E satisfies $m(E) = \inf\{m(U) : U \supset E \text{ open}\}$, and for bounded E, $m(E) = \sup\{m(K) : K \subset E \text{ compact}\}$ (apply the outer approximation to $[a, b] - E$). Not every set is measurable (Vitali): choosing one point in $[0, 1)$ from each coset of $\mathbb{Q}$ gives a set V whose rational translates mod 1 are disjoint and cover $[0, 1)$, so $m(V)$ could be neither 0 nor positive.

10.2 Measurable functions and the integral

Motivation. With measure in hand, we integrate simple functions and then approximate. Measurable functions are closed under all limit operations, so this class is large enough for analysis.

Let $E \in \mathfrak{M}$ and $\mathbb{R}^* = [-\infty, \infty]$. $f : E \to \mathbb{R}^*$ is **measurable** if $\{x \in E : f(x) > a\} \in \mathfrak{M}$ for every $a \in \mathbb{R}$; a complex function is measurable if its real and imaginary parts are.

Proposition 10.5 (10.2.1)

For $f : E \to \mathbb{R}^*$ the following are equivalent:
(a) f is measurable;
(b) $\{f \geq a\} \in \mathfrak{M}$ for all a;
(c) $\{f < a\} \in \mathfrak{M}$ for all a;
(d) $\{f \leq a\} \in \mathfrak{M}$ for all a.

Proof. $\{f \geq a\} = \bigcap_n \{f > a - 1/n\}$, $\{f > a\} = \bigcup_n \{f \geq a + 1/n\}$, and (c), (d) are the complements (relative to E) of (b), (a). ∎

Corollary 10.6 (10.2.2)

If f is measurable, then $\{f = a\}$ is measurable for every $a \in \mathbb{R}^*$.

Proof. $\{f = a\} = \{f \geq a\} \cap \{f \leq a\}$ for real a, and $\{f = \infty\} = \bigcap_n \{f \geq n\}$. ∎

Proposition 10.7 (10.2.3)

(a) Continuous $f : E \to \mathbb{R}$ are measurable.
(b) If f, g are measurable and real-valued and $a \in \mathbb{R}$, then $f + g$, fg, af are measurable.
(c) If each f_n is measurable, so are $\inf_n f_n$, $\sup_n f_n$, $\liminf f_n$, $\limsup f_n$, and hence $\lim f_n$ where it exists.

Proof. (a) $\{f > a\} = U \cap E$ with U open.

(b) $\{f + g > a\} = \bigcup_{r \in \mathbb{Q}} \{f > r\} \cap \{g > a - r\}$; f^2 is measurable since $\{f^2 > a\} = \{f > \sqrt{a}\} \cup \{f < -\sqrt{a}\}$ for $a \geq 0$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

(c) $\{\sup f_n > a\} = \bigcup \{f_n > a\}$, $\inf f_n = -\sup(-f_n)$, $\limsup f_n = \inf_n \sup_{k \geq n} f_k$. ∎

A **simple function** is $s = \sum_{i=1}^{k} a_i \chi_{A_i}$ with $a_i \geq 0$ and $A_i \in \mathfrak{M}$; its integral is $\int_E s = \sum a_i m(A_i \cap E)$ (independent of the representation, by passing to a common disjoint refinement and using additivity). For measurable $f : E \to [0, \infty]$,

$$\int_E f = \sup\Big\{ \int_E s : 0 \leq s \leq f,\ s \text{ simple} \Big\}.$$

For simple functions the two definitions agree, and $0 \leq f \leq g$ implies $\int f \leq \int g$.

Proposition 10.8 (10.2.4)

Every measurable $f : E \to [0, \infty]$ is the pointwise limit of an increasing sequence of simple functions.

Proof. Let $A_{n,i} = \{\frac{i-1}{2^n} \leq f < \frac{i}{2^n}\}$ $(1 \leq i \leq n2^n)$, $B_n = \{f \geq n\}$, and $s_n = \sum_i \frac{i-1}{2^n} \chi_{A_{n,i}} + n\chi_{B_n}$. Then $s_n \leq s_{n+1} \leq f$, and $f - s_n < 2^{-n}$ where $f < n$; also $s_n = n \to \infty$ where $f = \infty$. ∎

Theorem 10.9 (10.2.5)

If $A = \bigsqcup A_n$ with $A_n \in \mathfrak{M}$ disjoint and $f : A \to [0, \infty]$ is measurable, then $\int_A f = \sum_n \int_{A_n} f$.

Proof. For simple s, $\int_A s = \sum_n \int_{A_n} s$ by Theorem 10.4; so for simple $s \le f$, $\int_A s \le \sum \int_{A_n} f$, and taking the supremum, $\int_A f \le \sum \int_{A_n} f$. Conversely, given N and ϵ, choose simple $s_k \le f$ on A_k with $\int_{A_k} s_k > \int_{A_k} f - \epsilon/N$ $(k \le N)$; $s = \sum_{k\le N} s_k \chi_{A_k} \le f$, so $\int_A f \ge \sum_{k\le N} \int_{A_k} f - \epsilon$. ■

So $A \mapsto \int_A f$ is a measure. For measurable f, write $f^+ = \max(f, 0)$, $f^- = \max(-f, 0)$; $f \in L^1(E)$ if $\int_E |f| < \infty$, and then $\int_E f = \int_E f^+ - \int_E f^-$; for complex f, $\int f = \int \operatorname{Re} f + i \int \operatorname{Im} f$.

Corollary 10.10 (10.2.6)

Theorem 10.9 holds for every $f \in L^1(A)$, the series converging absolutely.

Proof. Apply Theorem 10.9 to $f^{\pm}$ (real and imaginary parts) and subtract; the series converge because $\sum \int_{A_n} |f| = \int_A |f| < \infty$. ■

Corollary 10.11 (10.2.7)

A measurable f is in $L^1(E)$ iff $|f| \in L^1(E)$, and then $|\int_E f| \le \int_E |f|$.

Proof. The first statement is the definition. Choose θ with $e^{-i\theta} \int f = |\int f|$; then $|\int f| = \int \operatorname{Re}(e^{-i\theta} f) \le \int |f|$ (using linearity, Theorem 10.14). ■

Proposition 10.12 (10.2.8)

If $f \ge 0$ is measurable and $\int_E f = 0$, then $f = 0$ almost everywhere.

Proof. With $A_n = \{f \ge 1/n\}$, $\frac{1}{n} m(A_n) \le \int f = 0$, so $m(\{f > 0\}) = m(\bigcup A_n) = 0$. ■

10.3 Convergence theorems

Motivation. The payoff of the Lebesgue integral: limits and integrals can be exchanged under very mild conditions (monotone convergence, Fatou, dominated convergence), with no uniform convergence required.

Theorem 10.13 (10.3.1, monotone convergence)

If $0 \le f_1 \le f_2 \le \cdots$ are measurable on E and $f = \lim f_n$, then $\int_E f = \lim \int_E f_n$.

Proof. f is measurable and $\int f_n \le \int f_{n+1} \le \int f$, so $L = \lim \int f_n \le \int f$. Let $0 \le s \le f$ be simple and $0 < c < 1$; $E_n = \{f_n \ge cs\}$ increase to E (where $f(x) = 0$, $x \in E_1$; otherwise $cs(x) < f(x)$). Then $\int f_n \ge \int_{E_n} f_n \ge c \int_{E_n} s \to c \int_E s$ (Theorem 10.9, continuity from below). So $L \ge c \int s$; let $c \to 1$ and take the supremum over s. ■

Theorem 10.14 (10.3.2, linearity)

(a) If $f, g \in L^1(E)$, then $f + g \in L^1(E)$; if $f, g \ge 0$ or $f, g \in L^1(E)$, then $\int (f + g) = \int f + \int g$.
(b) If $f \in L^1(E)$ and $a \in \mathbb{C}$, then $af \in L^1(E)$ and $\int af = a \int f$.

Proof. For simple functions this is immediate. For $f, g \ge 0$ take increasing simple $s_n \to f$, $t_n \to g$ (Proposition 10.8); $s_n + t_n \uparrow f + g$, and Theorem 10.13 gives the formula. Then $|f + g| \le |f| + |g|$

gives integrability. For real $f, g \in L^1$ and $h = f + g$, $h^+ + f^- + g^- = h^- + f^+ + g^+$; integrate and rearrange. Complex functions: real and imaginary parts.

(b) is similar (for $a = i$ the parts are exchanged). ■

Corollary 10.15 (10.3.3)

If $f_n : E \to [0, \infty]$ are measurable, then $\int_E \sum_n f_n = \sum_n \int_E f_n$.

Proof. Apply Theorem 10.13 to the partial sums, using Theorem 10.14. ■

Theorem 10.16 (10.3.4, Riemann and Lebesgue integrals)

Let $f : [a, b] \to \mathbb{R}$ be bounded.
(a) If f is Riemann integrable, then f is measurable and its Lebesgue integral equals its Riemann integral.
(b) f is Riemann integrable iff the set of its discontinuities has measure 0.

Proof. Choose partitions $P_1 \subset P_2 \subset \cdots$ with $\|P_k\| \to 0$, and let ℓ_k, u_k be the step functions equal to m_i, M_i on the open subintervals of P_k (and to f at partition points). Then $\ell_k \uparrow \ell$, $u_k \downarrow u$, $\ell \le f \le u$, and by monotone convergence (applied to $u_1 - u_k$ and $\ell_k - \ell_1$), $\int \ell = \lim L(f, P_k)$ and $\int u = \lim U(f, P_k)$. Let $P = \bigcup P_k$, a countable set. For $x \notin P$, $u(x)$ and $\ell(x)$ are the limits of sup and inf of f over shrinking neighbourhoods of x, so f is continuous at x iff $u(x) = \ell(x)$.

If f is Riemann integrable, then $U(f, P_k) - L(f, P_k) \to 0$ (proof of Theorem 5.8 of Volume I), so $\int (u - \ell) = 0$ and $u = \ell$ a.e. (Proposition 10.12). Then $f = u$ a.e.; f is measurable (Theorem 10.3(e)) and $\int f = \int u = \lim U(f, P_k)$ is the Riemann integral. This proves (a) and, with the characterization of continuity, the necessity in (b). Conversely, if the discontinuities have measure 0, then $u = \ell$ a.e., so $\lim(U(f, P_k) - L(f, P_k)) = \int (u - \ell) = 0$, and f is Riemann integrable by Riemann's criterion. ■

Theorem 10.17 (10.3.5, Fatou's lemma)

If $f_n : E \to [0, \infty]$ are measurable, then $\int_E \liminf f_n \le \liminf \int_E f_n$.

Proof. $g_n = \inf_{k \ge n} f_k$ increases to $\liminf f_n$ and $\int g_n \le \int f_k$ for $k \ge n$; apply Theorem 10.13. ■

Theorem 10.18 (10.3.6, Lebesgue's dominated convergence theorem)

Let $f_n : E \to \mathbb{C}$ be measurable with (a) $|f_n| \le g$ for some $g \in L^1(E)$ and (b) $f(x) = \lim f_n(x)$ for each x. Then $f \in L^1(E)$ and $\lim \int_E |f_n - f| = 0$; in particular $\int f_n \to \int f$.

Proof. $|f| \le g$, so $f \in L^1$. Apply Fatou to $2g - |f_n - f| \ge 0$: $\int 2g \le \liminf \int (2g - |f_n - f|) = \int 2g - \limsup \int |f_n - f|$. Hence $\limsup \int |f_n - f| \le 0$. ■

The theorems remain valid when the hypotheses hold only almost everywhere, since sets of measure 0 do not affect integrals.

Theorem 10.19 (10.3.7, Beppo Levi)

If $f_n \in L^1(E)$ and $\sum \int_E |f_n| < \infty$, then $\sum f_n(x)$ converges absolutely for almost every x, its sum f is in $L^1(E)$, and $\int_E f = \sum \int_E f_n$.

Proof. $g = \sum |f_n|$ has $\int g = \sum \int |f_n| < \infty$ (Corollary 10.15), so $g < \infty$ a.e. There the series converges absolutely, with partial sums dominated by g; apply Theorem 10.18. ■

Theorem 10.20 (10.3.8, absolute continuity)

If $f \in L^1(E)$, then for every $\epsilon > 0$ there is $\delta > 0$ with $\int_A |f| < \epsilon$ for all measurable $A \subset E$ with $m(A) < \delta$.

Proof. $\min(|f|, n) \uparrow |f|$, so by monotone convergence choose N with $\int(|f| - \min(|f|, N)) < \epsilon/2$, and $\delta = \epsilon/(2N)$. Then $\int_A |f| \le \epsilon/2 + Nm(A) < \epsilon$. ∎

10.4 The spaces L^1 and L^2

Motivation. With the Lebesgue integral, L^1 and L^2 become complete. L^2 is a Hilbert space in which Fourier series converge in norm and Parseval's identity holds; this is the clean form of the Fourier theory of Chapter 9.

Identifying functions that agree a.e., $L^1(E)$ with $\|f\|_1 = \int |f|$ and $L^2(E)$ (measurable f with $\int |f|^2 < \infty$) with $\|f\|_2 = (\int |f|^2)^{1/2}$ are normed spaces; L^2 has the inner product $\langle f, g\rangle = \int f\bar{g}$ ($|f\bar{g}| \le \frac{1}{2}(|f|^2 + |g|^2)$).

Theorem 10.21 (10.4.1)

Every Cauchy sequence in $L^1(E)$ converges.

Proof. By Proposition 7.11 it suffices that $\sum \|f_n\|_1 < \infty$ implies convergence of $\sum f_n$ in L^1. By Theorem 10.19, $f = \sum f_n$ exists a.e. and is in L^1, and $\|f - \sum_{n \le N} f_n\|_1 \le \sum_{n>N} \|f_n\|_1 \to 0$. ∎

Theorem 10.22 (10.4.2)

$L^2(E)$ is a Banach space (a Hilbert space).

Proof. Let $\sum \|f_n\|_2 = M < \infty$ and $g = \sum |f_n|$. By the triangle inequality $\|\sum_{n \le N} |f_n|\|_2 \le M$, and monotone convergence gives $\int g^2 \le M^2$, so $g < \infty$ a.e. and $f = \sum f_n$ exists a.e. with $|f| \le g$. Since $|f - \sum_{n \le N} f_n|^2 \le g^2$ and tends to 0 a.e., dominated convergence gives $\|f - \sum_{n \le N} f_n\|_2 \to 0$. Apply Proposition 7.11. ∎

Theorem 10.23 (10.4.3, Riesz–Fischer)

For every $c \in \ell^2(\mathbb{Z})$ there is a periodic $f \in L^2[-\pi, \pi]$ with $\hat{f}(n) = c(n)$ for all n.

Proof. By orthonormality, $s_N = \sum_{|n| \le N} c(n) e^{inx}$ satisfies $\frac{1}{2\pi}\int |s_M - s_N|^2 = \sum_{N < |n| \le M} |c(n)|^2$, so (s_N) is Cauchy in L^2 and converges to some f (Theorem 10.22). By Cauchy–Schwarz, $\hat{f}(n) = \lim_N \widehat{s_N}(n) = c(n)$. ∎

Thus $f \mapsto \hat{f}$ is an isometric isomorphism of $L^2[-\pi, \pi]$ (with $\frac{1}{2\pi}\int$) onto $\ell^2(\mathbb{Z})$, once injectivity is known (Theorem 10.26).

Proposition 10.24 (10.4.4)

If $f \in L^1(E)$ and $\int_A f = 0$ for every measurable $A \subset E$, then $f = 0$ a.e.

Proof. With $A = \{\operatorname{Re} f > 0\}$, $\int_A \operatorname{Re} f = \operatorname{Re} \int_A f = 0$, so $\operatorname{Re} f \chi_A = 0$ a.e. (Proposition 10.12) and $m(A) = 0$. Similarly for $\{\operatorname{Re} f < 0\}$, $\{\operatorname{Im} f > 0\}$, $\{\operatorname{Im} f < 0\}$. ∎

Lemma 10.25 (10.4.5)

For every measurable $E \subset [a,b]$ there are continuous g_n with $|g_n| \le 1$ and $g_n \to \chi_E$ a.e.

Proof. By regularity choose compact K_n and open U_n with $K_n \subset E \subset U_n$ and $m(U_n - K_n) < 2^{-n}$, and let $g_n(x) = \frac{d(x,U_n^c)}{d(x,U_n^c)+d(x,K_n)}$, continuous, equal to 1 on K_n and 0 off U_n. Then $g_n(x) \ne \chi_E(x)$ only for $x \in U_n - K_n$. The set of x lying in infinitely many $U_n - K_n$ is contained in $\bigcup_{n \ge N}(U_n - K_n)$ for every N, so has measure at most 2^{1-N}, i.e. 0. Elsewhere $g_n(x) \to \chi_E(x)$. ■

Theorem 10.26 (10.4.6)

If $f \in L^1[-\pi,\pi]$ and $\hat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $f = 0$ a.e.

Proof. By hypothesis $\int fg = 0$ for every trigonometric polynomial g, hence for every continuous periodic g (Fejér, Theorem 9.11, gives uniformly convergent trigonometric polynomials; pass to the limit by dominated convergence). Let $A \subset [-\pi+\eta, \pi-\eta]$ be measurable. Lemma 10.25, with the U_n inside $(-\pi,\pi)$, gives continuous periodic g_n, $|g_n| \le 1$, $g_n \to \chi_A$ a.e.; by dominated convergence $\int_A f = \lim \int f g_n = 0$. Letting $\eta \to 0$, $\int_A f = 0$ for all measurable A, and Proposition 10.24 applies. ■

Hence two L^1 functions with the same Fourier coefficients agree a.e., and Parseval's identity holds for every $f \in L^2[-\pi,\pi]$.

Lemma 10.27 (10.4.7)

For $f \in L^2(E)$ and $\epsilon > 0$ there is a simple function s (a finite complex combination of characteristic functions of sets of finite measure) with $\|f - s\|_2 < \epsilon$.

Proof. For $f \ge 0$ take increasing simple $s_n \to f$ (Proposition 10.8); $|f - s_n|^2 \le f^2$, so $\|f - s_n\|_2 \to 0$ by dominated convergence, and $\int s_n^2 < \infty$ forces the sets in s_n to have finite measure. In general use $f = (\operatorname{Re} f)^+ - (\operatorname{Re} f)^- + i(\operatorname{Im} f)^+ - i(\operatorname{Im} f)^-$. ■

Theorem 10.28 (10.4.8)

For $f \in L^2[a,b]$ and $\epsilon > 0$ there is $g \in C[a,b]$ with $\|f - g\|_2 < \epsilon$.

Proof. By Lemma 10.27 it suffices to approximate χ_A, $A \subset [a,b]$ measurable. With g_n from Lemma 10.25, $|g_n - \chi_A|^2 \le 4$ and tends to 0 a.e., so $\|g_n - \chi_A\|_2 \to 0$ by dominated convergence on the finite interval. ■

Theorem 10.29 (10.4.9)

For $f \in L^1(E)$ and $\epsilon > 0$ there is a continuous g on $\mathbb{R}$ vanishing outside a bounded interval with $\|f - g\|_1 < \epsilon$ (norm taken over E).

Proof. By dominated convergence $\|f - f\chi_{[-N,N]}\|_1 \to 0$; approximate $f\chi_{[-N,N]}$ in L^1 by simple functions as in Lemma 10.27; approximate each $\chi_{A \cap E}$, $A \subset [-N,N]$, by g_n from Lemma 10.25 (with $U_n \subset (-N-1, N+1)$), using dominated convergence. ■

In particular L^1 and L^2 are the completions of the spaces $\mathcal{R}^1$, $\mathcal{R}^2$ of Chapter 7.

10.5 The Poisson summation formula

Motivation. The Fourier transform extends Fourier series to nonperiodic functions on $\mathbb{R}$. The Poisson summation formula connects the two and has applications from theta functions to sampling theory.

For $g \in L^1(\mathbb{R})$ let $G(x) = \sum_{n\in\mathbb{Z}} g(x + 2\pi n)$.

Theorem 10.30 (10.5.1)

The series defining G converges absolutely for almost every x, $G \in L^1[-\pi, \pi]$, and

$$\hat{G}(n) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(x)e^{-inx}\,dx, \qquad n \in \mathbb{Z}.$$

Proof. Integrals over $\mathbb{R}$ are translation invariant (Proposition 10.2(c) for sets, hence for simple functions and by monotone convergence for all $f \geq 0$). By Corollary 10.15, $\int_{-\pi}^{\pi}\sum_n |g(x+2\pi n)|\,dx = \sum_n \int_{-\pi+2\pi n}^{\pi+2\pi n}|g| = \int_{\mathbb{R}}|g| < \infty$. So the series converges absolutely a.e. and $G \in L^1$. By Theorem 10.19,

$$\hat{G}(n) = \frac{1}{2\pi}\sum_k \int_{-\pi}^{\pi} g(x + 2\pi k)e^{-inx}dx = \frac{1}{2\pi}\sum_k \int_{-\pi+2\pi k}^{\pi+2\pi k} g(y)e^{-iny}dy = \frac{1}{2\pi}\int_{\mathbb{R}} ge^{-iny}dy.$$

■

If moreover the Fourier series of G converges to $G(0)$ at 0 (for example if g is smooth and rapidly decreasing), this gives the **Poisson summation formula** $\sum_n g(2\pi n) = \frac{1}{2\pi}\sum_n \int_{\mathbb{R}} g(x)e^{-inx}dx$. Applied to $g(x) = e^{-ax^2}$ it yields the transformation formula for the theta function $\theta(x) = \sum_n e^{-\pi n^2 x}$: $\theta(1/x) = \sqrt{x}\,\theta(x)$.